Random events and their probabilities

Count the outcomes, count the favourable ones, and divide — but only when they are equally likely.

LESSON 79–80 | 2 × 40 MIN | ALGEBRA 11, §29 | IGX 24.2

LESSON CLOCK

15' | 20' | 20' | 20' | 5'

The sample space	15 min
The classical definition	20 min
Counting with combinations	20 min
Relative frequency	20 min
Homework	5 min

LEARNING OBJECTIVES

- Describe a trial by its sample space.
- Compute a probability from equally likely outcomes.
- Estimate a probability from relative frequency, and say when that is required.
- Use combinatorics to count outcomes that cannot be listed.

TEXTBOOK

National	pp. 342–352
Cambridge	Core & Extended, pp. 629–634
Workbook	Exercise 24.2

01 Explanation

The sample space

Grade 11 begins where Grade 10 left off, but the counting is harder: the sample spaces are now too big to list, and the combinatorics of Quarter III does the listing instead.

$$P(A) = \frac{m}{n},\ n = |U|,\ m \text{ the favourable count}$$

TRIAL	$\lvert U \rvert$	COUNTED BY
two dice	36	6^2
a five-card hand from 52	$2\,598\,960$	$C(52, 5)$
an arrangement of 5 books	120	$5!$
a three-digit code, digits repeatable	1000	10^3
a committee of 3 from 10	120	$C(10, 3)$

ORDER, OR NO ORDER?

A **hand** of cards, a **committee**, a **selection** — order does not matter, use $C(n, k)$. A **code**, an **arrangement**, a **podium** — order matters, use $P(n, k)$ or a power. Choosing the wrong one changes n by a factor of $k!$.

The classical definition in use

The pattern is always the same: count n, count m, divide — and use the **same counting method** for both.

EVENT	M	N	P
two dice total 7	6	36	$\frac{1}{6}$
a committee of 3 from 6 men and 4 women is all women	$C(4,3) = 4$	$C(10,3) = 120$	$\frac{1}{30}$
two aces in a five-card hand	$C(4,2)\cdot C(48,3)$	$C(52,5)$	≈ 0.0399
a random arrangement of $ABCDE$ starts with A	$4! = 24$	$5! = 120$	$\frac{1}{5}$

COUNT BOTH WITH THE SAME METHOD

If n is a number of **unordered** selections, m must be too. Mixing $C(52, 5)$ below with an ordered count above gives an answer wrong by $5!$ — a factor of 120.

Counting with combinations

$$C(n, k) = \frac{n!}{k!(n-k)!}, \quad P(n, k) = \frac{n!}{(n-k)!}$$

The standard shape of a selection problem: choose some from one group and the rest from another, then **multiply**.

Example. A committee of 4 from 7 men and 5 women, with exactly 2 women:

$$\frac{C(5, 2) \cdot C(7, 2)}{C(12, 4)} = \frac{10 \times 21}{495} = \frac{210}{495} = \frac{14}{33}$$

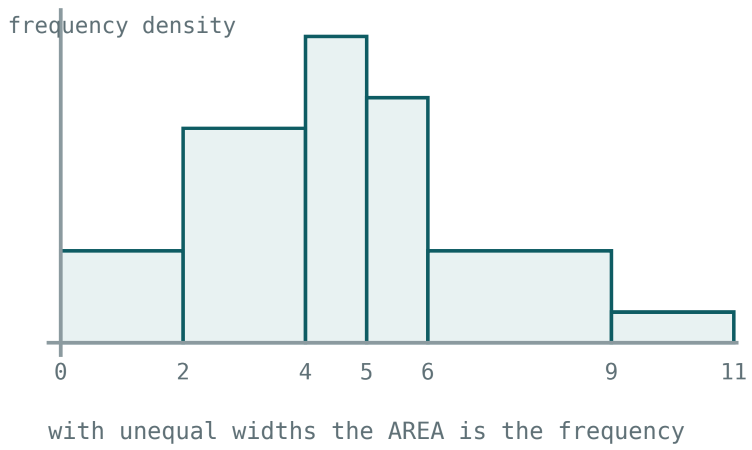
"AT LEAST" MEANS ADD THE CASES — OR USE THE COMPLEMENT

"At least one woman" on that committee: either add the cases with $1, 2, 3, 4$ women, or compute $1 - \frac{C(7, 4)}{C(12, 4)} = 1 - \frac{35}{495} = \frac{92}{99}$. The complement is one line instead of four.

Relative frequency

When outcomes are not equally likely, probability must be measured:

$$P(A) \approx \frac{\text{times A occurred}}{\text{number of trials}}$$



Frequencies from an experiment — the estimate that stands in for a count.

The estimate steadies as the number of trials grows. That is the **law of large numbers**, and it is the reason a table of experimental data can stand in for a theoretical probability at all.

SITUATION	WHICH DEFINITION
a fair die	classical — symmetry guarantees equal outcomes
a drawing pin	statistical — no symmetry to appeal to
a component failing in a year	statistical — from records
a point in a region	geometric — a ratio of measures

Expected number. In N trials with probability p, expect Np successes. In 500 throws of a fair die, expect ≈ 83 sixes — the number the binomial distribution of Lessons 86–89 will make precise.

02 Worked examples

EXAMPLE 1 **A committee of 3 is chosen from 6 men and 4 women. Find the probability that it is all women.**

1. $n = C(10, 3) = 120$
2. $m = C(4, 3) = 4$
3. $P = \dfrac{4}{120} = \dfrac{1}{30}$

ANSWER

$\dfrac{1}{30} \approx 0.033$

EXAMPLE 2 **A committee of 4 is chosen from 7 men and 5 women. Find the probability of exactly 2 women.**

1. $n = C(12, 4) = 495$

2. $m = C(5, 2)\cdot C(7, 2) = 10 \times 21 = 210$

   Choose from each group, then multiply.

3. $P = \dfrac{210}{495} = \dfrac{14}{33}$

ANSWER

$\dfrac{14}{33} \approx 0.424$

EXAMPLE 3 **The letters of *MATHS* are arranged at random. Find the probability that the arrangement begins with a vowel.**

1. $n = 5! = 120$

2. Only A is a vowel; the rest arrange in $4! = 24$ ways.

3. $P = \dfrac{24}{120} = \dfrac{1}{5}$

ANSWER

$\dfrac{1}{5}$

EXAMPLE 4 **A die is thrown 600 times and shows a six 118 times. Compare with the theoretical value.**

1. Relative frequency $\frac{118}{600} \approx 0.1967$
2. Theoretical $\frac{1}{6} \approx 0.1667$
3. Expected number $600 \times \frac{1}{6} = 100$

   The excess of 18 is within ordinary variation.

ANSWER

0.197 against 0.167; expected 100

03 Interactive model

Bring a pack of cards and deal ten hands of five; count the aces and compare with the computed probability.

Counting the sample space

A hand of 5 cards from 52 is counted by:

- 52^5
- $C(52, 5)$
- $P(52, 5)$
- $5!$

A 3-digit code with repeats:

- $C(10, 3)$
- 10^3
- $P(10, 3)$
- $3!$

Arrangements of 5 books:

- 5
- 25
- 120
- 32

A committee of 3 from 10:

- 120
- 720
- 1000
- 30

Two dice give:

- 12
- 21
- 36
- 6

For a bent drawing pin use:

- the classical definition
- relative frequency
- geometry
- nothing

Expected sixes in 600 throws:

- 60
- 100
- 120
- 600

“At least one” is quickest by:

- listing
- the complement
- a tree
- a table

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Random event	Tasodifiy hodisa	Случайное событие
Sample space	Elementar hodisalar fazosi	Пространство исходов
Equally likely	Teng imkoniyatli	Равновозможные
Favourable outcome	Qulay natija	Благоприятный исход
Relative frequency	Nisbiy chastota	Относительная частота
Combination	Kombinatsiya	Сочетание
Permutation	O‘rin almashtirish	Перестановка
Expected number	Kutilayotgan son	Ожидаемое число
Certain event	Muqarrar hodisa	Достоверное событие
Impossible event	Mumkin bo‘lmagan hodisa	Невозможное событие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The classical definition needs outcomes that are:

- few
- equally likely
- independent
- exclusive

A selection where order matters is counted by:

- $C(n, k)$
- $P(n, k)$
- n
- $k!$

m and n must be counted:

- any way
- by the same method
- by different methods
- approximately

The law of large numbers says the relative frequency:

- grows
- settles towards a fixed value
- oscillates for ever
- reaches 1

Choosing 2 from one group and 2 from another needs:

- addition
- multiplication
- subtraction
- division

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $C(10, 3)$

   ANSWER 120

2. $C(52, 2)$

   ANSWER 1326

3. $5!$

   ANSWER 120

4. $P(6, 2)$

   ANSWER 30

5. Outcomes for two dice

   ANSWER 36

6. P a die shows 4

   ANSWER $\frac{1}{6}$

7. Expected sixes in 600 throws

   ANSWER 100

Medium · the standard question

7 PROBLEMS

1. P a committee of 3 from 6 men and 4 women is all women

   ANSWER $\frac{1}{30}$

2. P exactly 2 women on a committee of 4 from 7 men and 5 women

   ANSWER $\frac{14}{33}$

3. P the arrangement of $MATHS$ begins with A

   ANSWER $\frac{1}{5}$

4. P two dice total 9

   ANSWER $\frac{1}{9}$

5. P a five-card hand is all hearts

   ANSWER $\frac{\mathrm{C}(13,5)}{\mathrm{C}(52,5)} \approx 0.000495$

6. P at least one woman on a committee of 4 from 7 men and 5 women

   ANSWER $\frac{92}{99}$

7. Relative frequency of 118 sixes in 600 throws

   ANSWER ≈ 0.197

Hard · stretch and reason

7 PROBLEMS

1. P a five-card hand contains exactly two aces

   ANSWER ≈ 0.0399

2. P a five-card hand is a full house

   ANSWER ≈ 0.00144

3. P the letters of *LEVEL* arrange to give *LEVEL*

   ANSWER $\frac{1}{30}$

4. From 10 people, P two named people sit together in a row

   ANSWER $\frac{1}{5}$

5. From 10 people in a circle, P two named people sit together

   ANSWER $\frac{2}{9}$

6. P a random 4-digit PIN has all different digits

   ANSWER $\frac{P(10,4)}{10^4} = 0.504$

7. A committee of 5 from 12 must contain a given person: P

   ANSWER $\frac{5}{12}$

07 Homework

Homework — 5 tasks

Write n and m on separate lines, each with the counting method named.

1. A committee of 3 is chosen from 5 men and 7 women. Find the probability that it contains exactly one man.
2. The letters of *NUMBER* are arranged at random. Find the probability that the arrangement begins and ends with a consonant.

3. Two cards are drawn from 52 without replacement. Find the probability that both are hearts.
4. A die is thrown 300 times and shows an even number 168 times. Find the relative frequency and compare it with the theoretical value.
5. Explain in two sentences why m and n must be counted by the same method.

Complementary events, operations on events and Euler–Venn diagrams

Events are sets; the diagram does the bookkeeping the algebra would otherwise have to.

LESSON 81–82 | 2 × 40 MIN | ALGEBRA 11, §30 | IGX 24.3

LESSON CLOCK

13' | 23' | 22' | 18' | 4'

Events as sets	13 min
The addition rule	23 min
Filling a Venn diagram	22 min
De Morgan and three sets	18 min
Homework	4 min

LEARNING OBJECTIVES

- Form the union, intersection and complement of events.
- Read and fill an Euler–Venn diagram from given totals.
- Use the addition rule, and recognise mutually exclusive events.
- Use De Morgan's laws to rewrite a compound event.

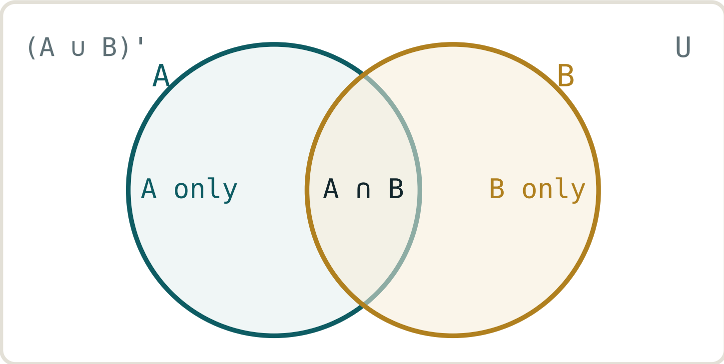
TEXTBOOK

National	pp. 353–364
Cambridge	Core & Extended, pp. 635–638
Workbook	Exercise 24.3

01 Explanation

Events as sets

IN WORDS	NOTATION	PROBABILITY
A or B	$A \cup B$	$P(A) + P(B) - P(A \cap B)$
A and B	$A \cap B$	from the data, or $P(A)P(B)$ if independent
not A	A'	$1 - P(A)$
A but not B	$A \cap B'$	$P(A) - P(A \cap B)$
neither	$(A \cup B)'$	$1 - P(A \cup B)$



$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Four regions, and every event in the table is one or two of them.

COMPLEMENTARY, EXCLUSIVE, EXHAUSTIVE

Two events are **complementary** when they are both exclusive ($A \cap B = \varnothing$) and exhaustive ($A \cup B = U$). Then and only then $P(B) = 1 - P(A)$. Exclusive alone is not enough.

The addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The subtraction removes the double count. For mutually exclusive events the overlap is empty and the rule shortens.

GIVEN	$P(A \cup B)$	$P(A \cap B)$
$P(A) = 0.6$, $P(B) = 0.3$	—	—
… exclusive	0.9	0
… independent	0.72	0.18
… with $P(A \cup B) = 0.8$	0.8	0.1

The last row is the standard examination move: the rule is used **backwards** to find the overlap from the union.

EXCLUSIVE AND INDEPENDENT ARE DIFFERENT, AND INCOMPATIBLE

Exclusive: $P(A \cap B) = 0$. Independent: $P(A \cap B) = P(A)P(B)$. If both probabilities are positive, the two cannot hold together — a question that says an event is “exclusive and independent” contains an error, or one of the probabilities is zero.

Filling a Venn diagram

Given totals, always fill the **overlap first**, then work outwards. Every region must be a count of things counted once.

Example. Of 40 pupils, 25 study physics, 18 chemistry, 10 both.

REGION	COUNT	ORDER TO FILL
$P \cap C$	10	first
P only	$25 - 10 = 15$	second
C only	$18 - 10 = 8$	third
neither	$40 - 33 = 7$	last

Now every probability is one division: $P(neither) = \frac{7}{40}$, $P(physics\ only) = \frac{15}{40} = \frac{3}{8}$, $P(P \mid C) = \frac{10}{18} = \frac{5}{9}$.

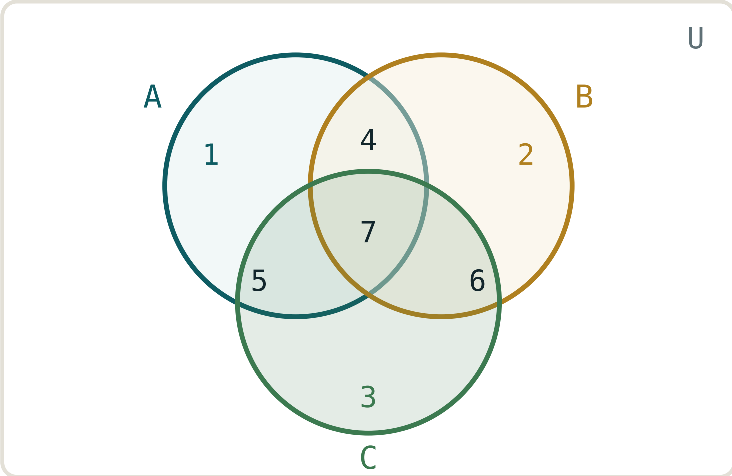
THE FOUR REGIONS MUST TOTAL $|U|$

$15 + 10 + 8 + 7 = 40$. Do this check every time; it catches the commonest error, which is writing 25 in the “physics only” region instead of 15.

De Morgan and three sets

$$(A \cup B)' = A' \cap B' \quad (A \cap B)' = A' \cup B'$$

In words: “not (A or B)” is “not A **and** not B”. The connective flips when the bar is distributed.



Three events, seven regions inside and one outside — the same filling method.

With three events the addition rule grows:

$$P(A \cup B \cup C) = \Sigma P(A) - \Sigma P(A \cap B) + P(A \cap B \cap C)$$

— add the singles, subtract the pairs, add the triple back. This is inclusion–exclusion, and the alternating signs continue for any number of events.

FILL THREE SETS FROM THE MIDDLE OUT

Start with $A \cap B \cap C$, then the three pairwise regions (each pair total **minus** the middle), then the three singles, then the outside. Any other order double-counts.

02 Worked examples

EXAMPLE 1 **Given $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cup B) = 0.8$, find $P(A \cap B)$ and $P(A \cap B')$.**

1. $0.8 = 0.6 + 0.3 - P(A \cap B)$
2. $P(A \cap B) = 0.1$
3. $P(A \cap B') = 0.6 - 0.1 = 0.5$

ANSWER

0.1 and 0.5

EXAMPLE 2 **Of 40 pupils, 25 study physics, 18 chemistry and 10 both. Find P(neither) and P(exactly one).**

1. Overlap 10; physics only 15; chemistry only 8.
2. Neither $40 - 33 = 7$.
3. $P(neither) = \dfrac{7}{40} = 0.175$
4. $P(exactly\ one) = \dfrac{23}{40} = 0.575$

ANSWER

0.175 and 0.575

EXAMPLE 3

Show that $(A \cup B)' = A' \cap B'$ on a Venn diagram, and use it to find P(neither) when $P(A \cup B) = 0.8$.

1. Shade outside both circles.
   That region is outside A and outside B.
2. $P((A \cup B)') = 1 - 0.8$
3. $= 0.2$

ANSWER

0.2

EXAMPLE 4

Of 60 students, 30 play football, 25 chess, 20 tennis; 12 football and chess, 8 chess and tennis, 10 football and tennis, 5 all three. How many play none?

1. $30 + 25 + 20 = 75$
   Singles.
2. $-(12 + 8 + 10) = -30$
   Pairs.
3. $+ 5 = 50$
   Triple back.
4. $60 - 50 = 10$

ANSWER

10 students

03 Interactive model

Draw the two circles on the board and let the class place the four numbers before any probability is computed.

Reading a Venn diagram

$P(A \cup B)$ equals:

- $P(A) + P(B)$
- $P(A) + P(B) - P(A \cap B)$
- $P(A)P(B)$
- $1 - P(A)$

$P(A')$ equals:

- $1 - P(A)$
- $P(A)$
- 0
- $1 + P(A)$

$(A \cup B)'$ equals:

- $A' \cup B'$
- $A' \cap B'$
- $A \cap B$
- $A \cup B$

Fill a Venn diagram starting from:

- the outside
- the overlap
- the largest region
- anywhere

Exclusive means:

- $P(A \cap B) = 0$
- $P(A \cap B) = P(A)P(B)$
- $P(A) = P(B)$
- $A = B$

For three events, the triple intersection is:

- subtracted
- added back
- ignored
- doubled

Complementary events are exclusive and:

- independent
- exhaustive
- equal
- disjointly independent

The four regions of a two-set diagram total:

- $|A| + |B|$
- $|U|$
- 1
- $|A \cap B|$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Union	Birlashma	Объединение
Intersection	Kesishma	Пересечение
Complement	To‘ldiruvchi	Дополнение
Euler–Venn diagram	Eyler–Venn diagrammasi	Диаграмма Эйлера–Венна
Mutually exclusive	Birgalikda bo‘lmagan	Несовместные
Exhaustive	To‘liq guruh	Полная группа
De Morgan’s laws	De Morgan qonunlari	Законы де Моргана
Universal set	Universal to‘plam	Универсальное множество
Difference of events	Hodisalar ayirmasi	Разность событий
Partition	Bo‘linish	Разбиение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The addition rule subtracts the overlap because:

- it is small
- it would be counted twice
- of De Morgan
- no reason

$(A \cap B)'$ equals:

- $A' \cap B'$
- $A' \cup B'$
- $A \cup B$
- $A \cap B$

Exclusive events with positive probability are:

- independent
- never independent
- complementary
- exhaustive

A Venn diagram is filled:

- left to right
- from the overlap outwards
- from the outside in
- randomly

For three events, pairs are:

- added
- subtracted
- ignored
- squared

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $P(A')$ when $P(A) = 0.35$

ANSWER 0.65

2. $P(A \cup B)$ when exclusive, $P(A) = 0.4$, $P(B) = 0.3$

ANSWER 0.7

3. $P(A \cap B)$ when independent, $P(A) = 0.5$, $P(B) = 0.4$

ANSWER 0.2

4. Write $(A \cup B)'$ another way

ANSWER $A' \cap B'$

5. Write $(A \cap B)'$ another way

ANSWER $A' \cup B'$

6. Of 40, 25 physics, 18 chemistry, 10 both: physics only

ANSWER 15

7. Same data: neither

ANSWER 7

Medium · the standard question

7 PROBLEMS

1. $P(A \cap B)$ when $P(A) = 0.6, P(B) = 0.3, P(A \cup B) = 0.8$

   ANSWER 0.1

2. Same: $P(A \cap B')$

   ANSWER 0.5

3. Same: P(neither)

   ANSWER 0.2

4. Of 40 pupils as above: P(exactly one subject)

   ANSWER 0.575

5. Same: $P(P \mid C)$

   ANSWER $\frac{5}{9}$

6. Are $P(A) = 0.5, P(B) = 0.4, P(A \cap B) = 0.2$ independent?

   ANSWER Yes

7. Of 60 with 30 F, 25 C, 20 T, 12 FC, 8 CT, 10 FT, 5 all: how many play none?

   ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. Of 60 as above: P(exactly two sports)

   ANSWER $\frac{15}{60} = \frac{1}{4}$

2. Of 60 as above: P(football only)

   ANSWER $\frac{13}{60}$

3. If $P(A) = 0.7$, $P(B) = 0.5$, find the range of $P(A \cap B)$

   ANSWER $0.2 \le P(A \cap B) \le 0.5$

4. If A and B are independent, show A and B' are too

   ANSWER $P(A \cap B') = P(A)(1 - P(B))$

5. Three events each of probability 0.5, pairwise independent, all three independent: $P(A \cup B \cup C)$

   ANSWER 0.875

6. Of 100 people, 60 read A, 50 read B, 40 read C, 25 A and B, 20 B and C, 15 A and C, 5 all: how many read none?

   ANSWER 5

7. Prove $P(A \cup B) \le P(A) + P(B)$ for any two events

   ANSWER $P(A \cap B) \ge 0$

07 Homework

Homework — 5 tasks

Draw the diagram and fill all four (or eight) regions before answering anything.

1. Given $P(A) = 0.55$, $P(B) = 0.4$ and $P(A \cup B) = 0.75$, find $P(A \cap B)$, $P(A \cap B')$ and P(neither).
2. Of 50 students, 32 study Russian, 28 English and 15 both. Draw the Venn diagram and find the probability that a student studies exactly one language.

3. State both De Morgan laws and illustrate each with a shaded Venn diagram.
4. Of 80 people, 40 own a car, 35 a bicycle, 30 a motorbike; 15 car and bicycle, 12 bicycle and motorbike, 10 car and motorbike, 5 all three. How many own none?
5. Explain in two sentences why mutually exclusive events with positive probabilities cannot be independent.

Home / Grade 11 / Algebra / Quarter 4

Addition and multiplication of probabilities

Two rules, one tree, and one question to ask first: does the second event depend on the first?

LESSON 83–85 | 3 × 40 MIN | ALGEBRA 11, §31 | IGX 24.1, 24.4

LESSON CLOCK

12' 24' 28' 28' 20' 8'

One question first	12 min
The multiplication rule	24 min
Trees	28 min
Total probability	28 min
Reversing the condition	20 min
Homework	8 min

LEARNING OBJECTIVES

- Use the multiplication rule for independent and for dependent events.
- Compute a conditional probability from data, a table or a tree.
- Build and use a tree diagram, including without replacement.
- Use the total probability formula, and reverse a conditional probability.

TEXTBOOK

National	pp. 365–382
Cambridge	Core & Extended, pp. 620–628, 639–642
Workbook	Exercise 24.4

01 Explanation

One question first

BEFORE ANY FORMULA

Does the second event depend on the first? If not, multiply the plain probabilities. If so, the second factor is a conditional probability.

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

When A and B are independent, $P(B \mid A) = P(B)$ and the rule collapses to $P(A)P(B)$. So the general rule always works; the independent one is a shortcut.

SITUATION	DEPENDENT?	SECOND FACTOR
two throws of a die	no	$\frac{1}{6}$
two cards, with replacement	no	$\frac{13}{52}$
two cards, without replacement	yes	$\frac{12}{51}$
two components from one batch	yes	from the reduced batch

The multiplication rule in use

$$P(A \cap B) = P(A)\,P(B \mid A) = P(B)\,P(A \mid B)$$

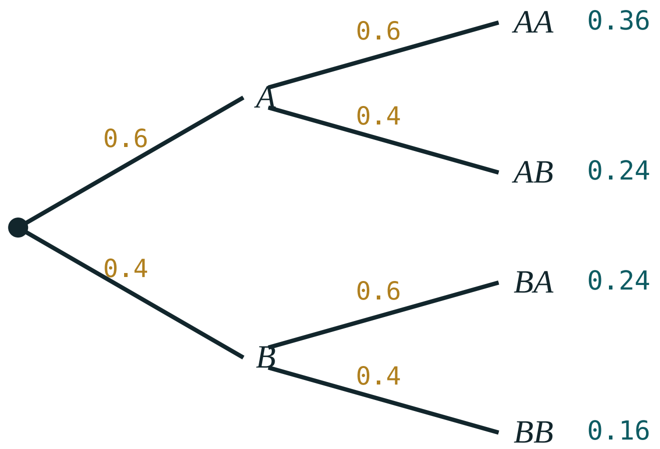
The rule can be entered from either end — a fact that will matter in the last section.

QUESTION	WORKING	ANSWER
two sixes in two throws	$\frac{1}{6} \times \frac{1}{6}$	$\frac{1}{36}$
two hearts, with replacement	$\frac{1}{4} \times \frac{1}{4}$	$\frac{1}{16}$
two hearts, without replacement	$\frac{13}{52} \times \frac{12}{51}$	$\frac{1}{17}$
two aces from 52	$\frac{4}{52} \times \frac{3}{51}$	$\frac{1}{221}$

THE DENOMINATOR DROPS AS WELL AS THE NUMERATOR

Without replacement, $\frac{13}{52}$ is followed by $\frac{12}{51}$ — **both** numbers change. Changing only the top is the standard error, and gives an answer about *2\%* too large.

Trees



Multiply along a branch, add down the ends — the two rules in one picture.

THE TWO TREE RULES

- **Along** a branch: multiply (that is the multiplication rule).
- **Down** the ends: add (that is the addition rule, on exclusive paths).

Every fork's probabilities total 1; every set of end-probabilities totals 1. Two free checks.

Worked case. A bag holds 5 red and 3 blue balls; two are drawn without replacement.

PATH	PRODUCT	VALUE
RR	$\frac{5}{8} \times \frac{4}{7}$	$\frac{20}{56}$
RB	$\frac{5}{8} \times \frac{3}{7}$	$\frac{15}{56}$
BR	$\frac{3}{8} \times \frac{5}{7}$	$\frac{15}{56}$
BB	$\frac{3}{8} \times \frac{2}{7}$	$\frac{6}{56}$

They total $\frac{56}{56} = 1$. So $P(same) = \frac{26}{56} = \frac{13}{28}$ and $P(different) = \frac{30}{56} = \frac{15}{28}$.

Total probability

When the first stage splits the sample space into exclusive, exhaustive cases H_1, H_2, …, the probability of a later event is the sum over the branches:

$$P(A) = P(H_1)P(A \mid H_1) + P(H_2)P(A \mid H_2) + \dots$$

This is exactly “add down the ends”, written as a formula.

Example. Two factories supply lamps: 60% from F_1 with 2% defective, 40% from F_2 with 5% defective. The probability that a random lamp is defective:

$$P(D) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.020 = 0.032$$

THE BRANCHES MUST PARTITION

The cases must be mutually exclusive and cover everything: $P(H_1) + P(H_2) + \dots = 1$. If they do not, the formula gives a number that is not a probability — often greater than 1, which is the check.

Reversing the condition

Total probability answers “how likely is the effect?”. The reverse question — “given the effect, how likely is each cause?” — uses the same numbers the other way round:

$$P(H_1 \mid A) = \frac{\mathrm{P(H_1)\,P(A \mid H_1)}}{\mathrm{P(A)}}$$

The lamp again. A lamp is found defective; the probability it came from F_1:

$$\frac{0.012}{0.032} = 0.375$$

So although F_1 makes 60% of the lamps, it accounts for only 37.5% of the defective ones — because its defect rate is lower.

$P(A \mid B)$ AND $P(B \mid A)$ ARE DIFFERENT NUMBERS

Here $P(D \mid F_1) = 0.02$ but $P(F_1 \mid D) = 0.375$ — a factor of nearly 19. Swapping the two is the commonest and most consequential error in probability, in examinations and outside them.

02 Worked examples

EXAMPLE 1 **Two cards are drawn from 52 without replacement. Find P(both hearts).**

1. $P(first) = \frac{13}{52} = \frac{1}{4}$

2. $P(second \mid first) = \frac{12}{51}$

   Both numbers change.

3. $P = \frac{1}{4} \times \frac{12}{51} = \frac{1}{17}$

ANSWER

$\frac{1}{17} \approx 0.059$

EXAMPLE 2 **A bag holds 5 red and 3 blue balls; two are drawn without replacement. Find P(same colour).**

1. $P(RR) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$

2. $P(BB) = \frac{3}{8} \times \frac{2}{7} = \frac{6}{56}$

3. $P = \frac{26}{56} = \frac{13}{28}$

   Add down the ends.

ANSWER

$\frac{13}{28} \approx 0.464$

EXAMPLE 3 **Factory F_1 makes 60% of lamps with 2% defective; F_2 makes 40% with 5%. Find the probability that a lamp is defective.**

1. $P(D \mid F_1) = 0.02,\ P(D \mid F_2) = 0.05$
2. $P(D) = 0.6(0.02) + 0.4(0.05)$
3. $= 0.012 + 0.020 = 0.032$

ANSWER

0.032

EXAMPLE 4 **A lamp is defective. Find the probability that it came from F_1.**

1. $P(F_1 \cap D) = 0.012$
2. $P(D) = 0.032$

   From the previous example.
3. $P(F_1 \mid D) = \dfrac{0.012}{0.032} = 0.375$

ANSWER

0.375

03 Interactive model

Draw the lamp tree on the board and label all four end-probabilities; the reversal is then a single fraction the class can see.

Along the branch, down the ends

Along a branch you:

- add
- multiply
- subtract
- divide

Down the ends you:

- add
- multiply
- subtract
- divide

Without replacement, the second denominator:

- stays
- drops by one
- doubles
- is 1

$P(A \cap B)$ in general:

- $P(A)P(B)$
- $P(A)P(B \mid A)$
- $P(A) + P(B)$
- $P(B \mid A)$

The branches of total probability must:

- be equal
- partition the space
- be independent
- be few

All end-probabilities of a tree total:

- 0
- 1
- 2
- anything

$P(D \mid F_1) = 0.02$ and $P(F_1 \mid D) = 0.375$ shows:

- an error
- the two are different
- they are equal
- nothing

For independent events, $P(B \mid A)$ equals:

- $P(A)$
- $P(B)$
- 0
- 1

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Multiplication rule	Ko‘paytirish qoidasi	Правило умножения
Conditional probability	Shartli ehtimollik	Условная вероятность
Independent events	Bog‘liqsiz hodisalar	Независимые события
Dependent events	Bog‘liq hodisalar	Зависимые события
Tree diagram	Daraxt diagrammasi	Дерево исходов
With replacement	Qaytarib	С возвращением
Without replacement	Qaytarmasdan	Без возвращения
Total probability	To‘la ehtimollik	Полная вероятность
Bayes’ rule	Bayes formulasi	Формула Байеса
Partition	To‘liq guruh	Полная группа

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The general multiplication rule is:

- $P(A)P(B)$
- $P(A)P(B \mid A)$
- $P(A) + P(B)$
- $P(A) - P(B)$

Without replacement makes the events:

- independent
- dependent
- exclusive
- certain

Total probability is the formula for:

- along a branch
- down the ends
- the complement
- the union

Reversing a conditional probability uses:

- the addition rule
- the same joint probability, divided differently
- independence
- a new experiment

A tree fork's probabilities total:

- 0
- 1
- $|U|$
- anything

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up — 7 PROBLEMS

1. P two sixes in two throws

   ANSWER $\frac{1}{36}$

2. P two hearts, with replacement

   ANSWER $\frac{1}{16}$

3. P two hearts, without replacement

   ANSWER $\frac{1}{17}$

4. P two aces from 52 without replacement

   ANSWER $\frac{1}{221}$

5. Bag of 5 red, 3 blue: $P(RR)$ without replacement

   ANSWER $\frac{5}{14}$

6. Same bag: $P(BB)$

   ANSWER $\frac{3}{28}$

7. Second factor after drawing one heart from 52

   ANSWER $\frac{12}{51}$

Medium · the standard question

7 PROBLEMS

1. Bag of 5 red, 3 blue: P(same colour)

   ANSWER $\frac{13}{28}$

2. Same bag: P(different colours)

   ANSWER $\frac{15}{28}$

3. Lamps: $P(D)$ with $0.6/0.02$ and $0.4/0.05$

   ANSWER 0.032

4. Same: $P(F_1 \mid D)$

   ANSWER 0.375

5. Same: $P(F_2 \mid D)$

   ANSWER 0.625

6. Three cards from 52 without replacement: P all hearts

   ANSWER $\frac{11}{850} \approx 0.0129$

7. $P(B \mid A)$ when $P(A) = 0.4$, $P(A \cap B) = 0.1$

   ANSWER 0.25

Hard · stretch and reason

7 PROBLEMS

1. A bag of 4 white, 6 black: P the second drawn is white

   ANSWER $\frac{2}{5}$

2. Two bags: I has 3R 2B, II has 1R 4B. A bag is chosen at random, then a ball: $P(R)$

   ANSWER 0.4

3. Same: P(bag I | red drawn)

   ANSWER 0.75

4. A test is 95% accurate; 1% of people have the disease. P(disease | positive)

   ANSWER ≈ 0.161

5. A die is thrown until a six appears: P it takes exactly three throws

   ANSWER $\frac{25}{216}$

6. Three components each fail with probability 0.1, independently: P at least one fails

   ANSWER 0.271

7. Two cards from 52 without replacement: P at least one is an ace

   ANSWER $\frac{33}{221} \approx 0.149$

07 Homework

Homework — 5 tasks

Draw the tree for every multi-stage question; label every branch before multiplying.

1. A bag holds 6 green and 4 yellow counters; two are drawn without replacement. Find the probabilities of two greens, of two yellows and of one of each.
2. Two cards are drawn from 52 without replacement. Find the probability that at least one is a king.

3. Machine A makes $70\backslash\%$ of the output with $3\backslash\%$ faulty; machine B makes $30\backslash\%$ with $6\backslash\%$. Find the probability that an item is faulty, and the probability it came from B given that it is.
4. A die is thrown three times. Find the probability of at least one six.
5. Explain in two sentences, with the lamp numbers, why $P(A \mid B)$ and $P(B \mid A)$ are different.

The binomial and normal distributions

Counting successes in n trials, and the bell curve that takes over when n is large.

LESSON 86–89 | 4 × 40 MIN | ALGEBRA 11, §32 | PROBABILITY & STATISTICS 1

LESSON CLOCK

20' | 35' | 30' | 35' | 35' | 5

When a situation is binomial	20 min
The formula	35 min
Mean and variance	30 min
The normal curve	35 min
Standardising	35 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise a binomial situation and state its two parameters.
- Compute P(X = k), the mean and the variance of a binomial distribution.
- Use the normal curve and the 68–95–99.7 rule.
- Standardise a value with z = (x − μ)/σ and read a probability from it.

TEXTBOOK

National	pp. 383–406
Cambridge	P&S 1, pp. 108–150
Workbook	Exercise S1–S3

01 Explanation

When a situation is binomial

FOUR CONDITIONS, ALL NEEDED

1. a **fixed** number n of trials;
2. each trial has exactly **two** outcomes, success or failure;
3. the probability p of success is the **same** every time;
4. the trials are **independent**.

Then X, the number of successes, is **binomial**: $X \sim B(n, p)$.

SITUATION	BINOMIAL?	WHY
10 throws of a die, counting sixes	yes	$B(10, \frac{1}{6})$
20 components tested, *3\%* faulty	yes	$B(20, 0.03)$
cards drawn **without** replacement	no	p changes
throwing until the first six	no	n is not fixed

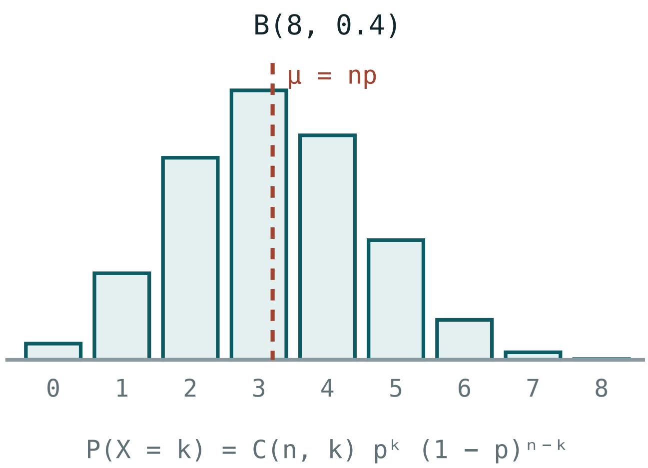
“WITHOUT REPLACEMENT” BREAKS CONDITION 3

Drawing without replacement changes p at every step, so the count is **not** binomial. Sampling from a very large population is close enough to be treated as binomial — but say so.

The formula

$$P(X = k) = C(n, k)\, p^k (1 - p)^{n-k}, \quad k = 0, 1, \ldots, n$$

Read it as three factors: **how many ways** k successes can be arranged, times the probability of k successes, times the probability of $n - k$ failures.



B(8, 0.4) — the whole distribution, with the mean marked.

Example. $X \sim B(5, 0.2)$:

K	$C(5, K)$	$P(X = K)$
0	1	0.328
1	5	0.410
2	10	0.205
3	10	0.051
4	5	0.006
5	1	0.0003

THE PROBABILITIES TOTAL 1

They are the terms of $(p + (1 - p))^n$ expanded by the binomial theorem of Lesson 101 — which is where the distribution takes its name. So the column always sums to 1, and that is the check.

Mean and variance

$$\mu = E(X) = np \quad \sigma^2 = Var(X) = np(1 - p) \quad \sigma = \sqrt{np(1 - p)}$$

DISTRIBUTION	M	Σ^2	Σ
$B(10, \frac{1}{2})$	5	2.5	1.58
$B(60, \frac{1}{6})$	10	8.33	2.89
$B(100, 0.03)$	3	2.91	1.71

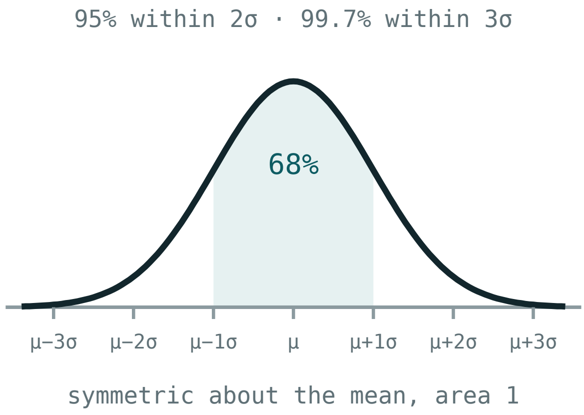
The mean is exactly the “expected number” of Lesson 79. The variance is largest at $p = 0.5$ and shrinks towards 0 as p approaches 0 or 1 — a near-certain outcome varies little.

M NEED NOT BE A POSSIBLE VALUE

$B(5, 0.2)$ has $\mu = 1$, but $B(7, 0.5)$ has $\mu = 3.5$ — and X is always a whole number. The mean is an average over many repetitions, not a prediction of one.

The normal curve

Many measured quantities — heights, errors, yields — cluster symmetrically about a mean. Their distribution is the **normal** one, $N(\mu, \sigma^2)$, drawn as the bell curve.



Symmetric about μ, with almost all the area within three standard deviations.

THE 68–95–99.7 RULE

> $P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$ $P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$ $P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$

Three numbers that answer most questions without a table.

Example. Heights are $N(170, 8^2)$ cm. Then about 68% of people are between 162 and 178 cm, about 95% between 154 and 186, and a height above 194 — three standard deviations up — occurs in about 0.15% of people.

Why it matters here. When n is large, the binomial bars themselves take the shape of the normal curve, with $\mu = np$ and $\sigma = \sqrt{np(1-p)}$. That is why the two distributions belong in one lesson.

Standardising

Every normal distribution is the same curve, re-scaled. Convert a value to its **z-score** — how many standard deviations it lies from the mean:

$$z = \frac{x - \mu}{\sigma}$$

DATA	Z	MEANING
$x = 178$ in $N(170, 8^2)$	1	one σ above the mean
$x = 154$ in $N(170, 8^2)$	-2	two σ below
$x = 62$ in $N(50, 6^2)$	2	top $2.5\backslash\%$ roughly

Once standardised, values from different distributions can be compared: a mark of 62 in a test with $N(50, 6^2)$ ($z = 2$) beats a mark of 78 in one with $N(70, 10^2)$ ($z = 0.8$), even though the raw score is lower.

SYMMETRY HALVES THE WORK

$P(Z < -z) = P(Z > z) = 1 - P(Z < z)$. A table of positive z is therefore enough for every question, and $P(Z < 0) = 0.5$ exactly.

02 Worked examples

EXAMPLE 1 **A die is thrown 5 times. Find the probability of exactly two sixes.**

1. $X \sim B(5, \frac{1}{6})$
2. $P(X = 2) = C(5, 2)(\frac{1}{6})^2(\frac{5}{6})^3$
3. $= 10 \times \frac{1}{36} \times \frac{125}{216}$
4. $= \frac{1250}{7776} \approx 0.161$

ANSWER

≈ 0.161

EXAMPLE 2 **For $X \sim B(20, 0.3)$, find μ, σ and $P(X \geq 1)$.**

1. $\mu = 20(0.3) = 6$

2. $\sigma^2 = 20(0.3)(0.7) = 4.2 \Rightarrow \sigma \approx 2.05$

3. $P(X = 0) = 0.7^{20} \approx 0.000798$

4. $P(X \geq 1) \approx 0.9992$

   Complement.

ANSWER

$\mu = 6$, $\sigma \approx 2.05$, $P(X \geq 1) \approx 0.999$

EXAMPLE 3 **Heights are $N(170, 8^2)$ cm. Find the proportion between 162 and 186 cm.**

1. $$z_1 = \frac{162 - 170}{8} = -1$$

2. $$z_2 = \frac{186 - 170}{8} = 2$$

3. $P(-1 < Z < 2) = 0.9772 - 0.1587$

4. ≈ 0.819

   About 82%.

ANSWER

$\approx 81.9\%$

EXAMPLE 4 **A mark of *62* in a test with $N(50, 6^2)$ and one of *78* in a test with $N(70, 10^2)$ — which is better?**

1. $z_1 = \dfrac{12}{6} = 2$

2. $z_2 = \dfrac{8}{10} = 0.8$

3. The first is two standard deviations above; the second only 0.8.

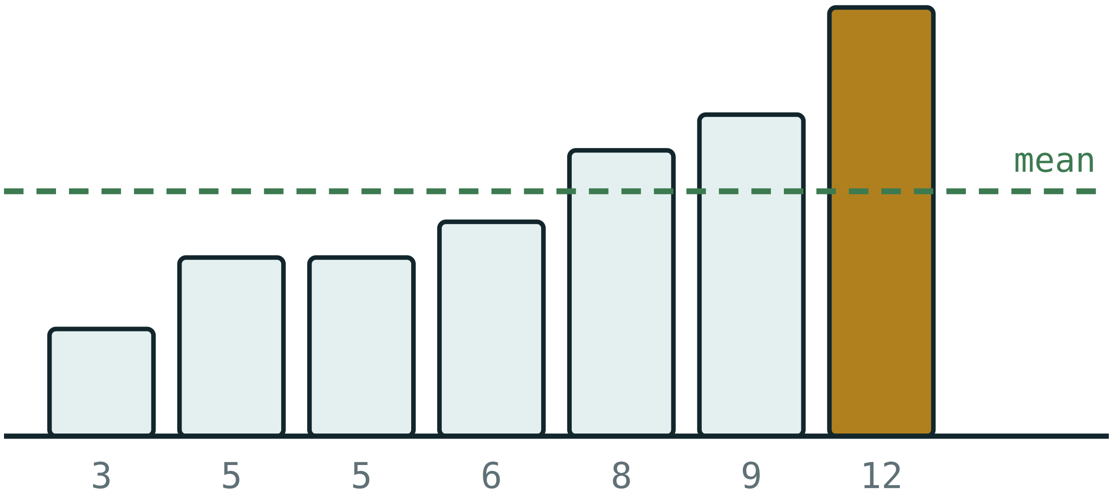
ANSWER

The mark of 62 — a higher z-score

03 Interactive model

Toss ten coins thirty times and tally the number of heads; the class builds B(10, 0.5) with its own hands, and the bell shape appears.

Mean, spread and the shape of the data



values `3, 5, 5, 6, 8, 9, 12` | mean `6.86` | median `6` | mode `5` | range `9`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Random variable	Tasodifiy miqdor	Случайная величина
Distribution	Taqsimot	Распределение
Binomial distribution	Binomial taqsimot	Биномиальное распределение
Trial	Sinov	Испытание
Success	Muvaffaqiyat	Успех
Mean (expectation)	Matematik kutilma	Математическое ожидание
Variance	Dispersiya	Дисперсия
Standard deviation	Standart chetlanish	Стандартное отклонение
Normal distribution	Normal taqsimot	Нормальное распределение
Standardisation	Standartlashtirish	Стандартизация
z-score	z-baho	z-оценка
Symmetry of the curve	Egri chiziq simmetriyasi	Симметрия кривой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A binomial distribution needs:

- a fixed n and constant p
- a large n
- a small p
- nothing

Drawing without replacement is:

- binomial
- not binomial
- normal
- geometric

$P(X = k)$ equals:

- p^k
- $C(n, k)p^k(1 - p)^{(n-k)}$
- np
- $C(n, k)$

The mean of $B(n, p)$:

- np
- $np(1 - p)$
- p
- n

The variance of $B(n, p)$:

- np
- $np(1 - p)$
- $\sqrt{\mathrm{np}}$
- n

Within two standard deviations lies about:

- 68%
- 95%
- 99.7%
- 50%

z equals:

- $\frac{x - \mu}{\sigma}$
- $\frac{\mu - x}{\sigma}$
- $x\sigma + \mu$
- $\frac{x}{\sigma}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. μ of $B(10, 0.5)$

ANSWER 5

2. μ of $B(60, \frac{1}{6})$

ANSWER 10

3. σ^2 of $B(10, 0.5)$

ANSWER 2.5

4. σ of $B(100, 0.03)$

ANSWER ≈ 1.71

5. $P(X = 0)$ for $B(5, 0.2)$

ANSWER 0.328

6. Is drawing without replacement binomial?

ANSWER No

7. z for $x = 178$ in $N(170, 8^2)$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. P exactly two sixes in five throws

   ANSWER ≈ 0.161

2. $P(X = 2)$ for $B(5, 0.2)$

   ANSWER 0.205

3. $P(X \geq 1)$ for $B(20, 0.3)$

   ANSWER ≈ 0.9992

4. μ and σ for $B(20, 0.3)$

   ANSWER $6, \approx 2.05$

5. Proportion within one σ of the mean

   ANSWER $\approx 68\backslash\%$

6. Proportion of $N(170, 8^2)$ between 154 and 186

   ANSWER $\approx 95\backslash\%$

7. z for $x = 62$ in $N(50, 6^2)$

   ANSWER 2

Hard · stretch and reason

7 PROBLEMS

1. P at least three sixes in ten throws of a die

   ANSWER ≈ 0.225

2. For $B(n, 0.5)$ with $\sigma = 2$, find n

   ANSWER $n = 16$

3. Proportion of $N(170, 8^2)$ between 162 and 186

   ANSWER $\approx 81.9\%$

4. The most likely value of $B(20, 0.3)$

   ANSWER $k = 6$

5. A test is passed by 80%; P exactly 8 of 10 pass

   ANSWER ≈ 0.302

6. Compare 62 in $N(50, 6^2)$ with 78 in $N(70, 10^2)$

   ANSWER The first — $z = 2$ against 0.8

7. For $B(n, p)$ with $\mu = 12$ and $\sigma^2 = 4.8$, find n and p

   ANSWER $n = 20, p = 0.6$

07 Homework

Homework — 5 tasks

State n and p — or μ and σ — before any calculation.

1. A coin is thrown 8 times. Find the probability of exactly 5 heads, and the mean and standard deviation of the number of heads.
2. A component is faulty with probability 0.04. In a batch of 25, find the probability of no faulty components and of at least two.
3. Test marks are $N(60, 12^2)$. Find the proportion of marks above 84 and between 48 and 72.
4. For $X \sim B(n, 0.25)$ with mean 5, find n, σ and $P(X = 5)$.

5. Explain in two sentences why sampling without replacement from a very large population may still be treated as binomial.

Control work 7, and work on the mistakes

Probability and distributions in one paper, then the block drawn as a decision tree.

LESSON 90–91 | 2 × 40 MIN | ALGEBRA 11, NAZORAT ISHI 7 | P&S 1 REVIEW

LESSON CLOCK

Instructions	3 min
The paper	42 min
Answers	11 min
Diagnosis and rewrite	19 min
The decision tree	5 min

LEARNING OBJECTIVES

- Apply the counting, addition and multiplication rules under time.
- Compute a conditional probability with the right denominator.
- Use the binomial formula and the normal 68–95–99.7 rule.
- Name each lost mark and rewrite the solution in full.

TEXTBOOK

National	pp. 407–410
Cambridge	P&S 1, pp. 151–154
Workbook	Control paper A7

01 Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	FROM
1	A committee of 4 from 6 men and 5 women: P(exactly 2 women) and P(at least 1 woman)	*7*	L79–80
2	Given $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$: find $P(A \cap B)$, $P(A \cap B')$ and say whether the events are independent	*6*	L81–82
3	A bag of 7 red and 5 blue, two drawn without replacement: the tree, P(same colour), P(second red)	*7*	L83–85
4	Two machines, $65/35\backslash\%$ of output with $2/6\backslash\%$ faulty: P(faulty) and P(machine B | faulty)	*7*	L83–85
5	$X \sim B(12, 0.25)$: find μ, σ, $P(X = 3)$ and $P(X \geq 1)$	*7*	L86–89
6	Masses are $N(500, 15^2)$ g: the proportion above 530 g, and between 485 and 515 g	*6*	L86–89

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for the complement rather than four cases; Q3 one for changing the second fork; Q4 one for the denominator $P(faulty)$ rather than 1; Q5 one for the complement in $P(X \geq 1)$. Four of the forty marks are for one decision each.

The decision tree for the whole block

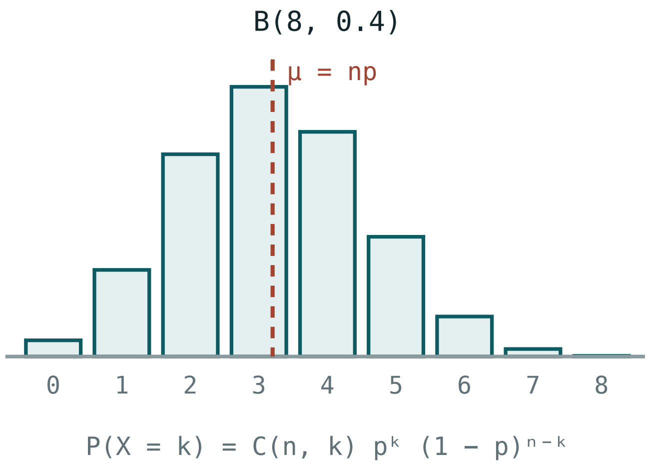
THE QUESTION SAYS	USE	FORMULA
“committee”, “hand”, “selection”	combinations	$\frac{C(a, j)\cdot C(b, k)}{C(a+b, j+k)}$
“or”, “either”	the addition rule	$P(A) + P(B) - P(A \cap B)$
“and”, “then”	the multiplication rule	$P(A)P(B \mid A)$
“at least one”	the complement	$1 - P(none)$
“given that”	conditional	$\frac{P(A \cap B)}{P(B)}$
“exactly k out of n”	binomial	$C(n, k)p\text{^}k(1-p)\text{^}(n-k)$
“a measured quantity”, “heights”, “masses”	normal	$z = \frac{x - \mu}{\sigma}$

THE FIVE SLIPS THAT ACCOUNT FOR MOST LOST MARKS

- counting m and n by different methods;
- adding without subtracting the overlap;
- keeping the denominator fixed in a without-replacement tree;
- swapping $P(A \mid B)$ and $P(B \mid A)$;
- treating a without-replacement count as binomial.

Looking forward

Lessons 92–100 are the Cambridge revision block: complex numbers and the Argand diagram, loci in the complex plane, and differential equations. It is the last new mathematics of the school course, and the first mathematics of the university one.



The last picture of the probability block — and the shape the normal curve takes over.

ONE HABIT TO CARRY FORWARD

Probability rewarded naming the situation before choosing a formula. Complex numbers and differential equations reward naming the **form** before choosing a method — Cartesian or polar, separable or not. The habit is the same; only the vocabulary changes.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: a committee of 4 from 6 men and 5 women — *P*(at least 1 woman).**

1. $n = C(11, 4) = 330$

2. $P(\text{no woman}) = \dfrac{C(6, 4)}{330} = \dfrac{15}{330}$

   The complement.

3. $P = 1 - \dfrac{15}{330} = \dfrac{315}{330} = \dfrac{21}{22}$

ANSWER

$\dfrac{21}{22} \approx 0.955$

EXAMPLE 2 **Model answer, Q4: machines 65\%/2\% and 35\%/6\%.**

1. $P(F) = 0.65(0.02) + 0.35(0.06)$

2. $= 0.013 + 0.021 = 0.034$

3. $P(B \mid F) = \dfrac{0.021}{0.034}$

4. ≈ 0.618

   Machine B makes a third of the output but most of the faults.

ANSWER

0.034 and ≈ 0.618

EXAMPLE 3 **Model answer, Q5: $X \sim B(12, 0.25)$.**

1. $\mu = 3$

2. $\sigma^2 = 12(0.25)(0.75) = 2.25 \Rightarrow \sigma = 1.5$

3. $P(X = 3) = C(12, 3)(0.25)^3(0.75)^9 \approx 0.258$

4. $P(X \geq 1) = 1 - 0.75^{12} \approx 0.968$

   Complement.

ANSWER

$\mu = 3, \sigma = 1.5, 0.258, 0.968$

03 Interactive model

Work Q4 twice — once dividing by 0.034, once by 1 — and let the class say which question each answers.

Twelve questions on the probability block

A committee of 3 from 10:

- 30
- 120
- 720
- 1000

Order matters means:

- $C(n, k)$
- $P(n, k)$
- $n!$
- $k!$

$P(A \cup B)$:

- $P(A) + P(B)$
- $P(A) + P(B) - P(A \cap B)$
- $P(A)P(B)$
- $1 - P(A)$

$(A \cup B)'$:

- $A' \cup B'$
- $A' \cap B'$
- $A \cap B$
- U

$P(A \cap B)$ in general:

- $P(A)P(B)$
- $P(A)P(B \mid A)$
- $P(A) + P(B)$
- 0

Without replacement, the denominator:

- stays
- drops by one
- doubles
- is 1

Total probability sums over:

- any events
- a partition
- independent events
- nothing

$P(B \mid A)$ divides by:

- $P(B)$
- $P(A)$
- 1
- $P(A \cup B)$

$B(n, p)$ needs p:

- small
- constant
- large
- variable

Its mean:

- np
- $np(1 - p)$
- p
- $\sqrt{np}$

Within two σ lies:

- 68\%
- 95\%
- 99.7\%
- 50\%

$z = 2$ means:

- twice the mean
- two σ above the mean
- a probability of 2
- nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Combination	Kombinatsiya	Сочетание
Conditional probability	Shartli ehtimollik	Условная вероятность
Total probability	To‘la ehtimollik	Полная вероятность
Binomial distribution	Binomial taqsimot	Биномиальное распределение
Standard deviation	Standart chetlanish	Стандартное отклонение
z-score	z-baho	z-оценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q1 is quickest by:

- four cases
- the complement
- a tree
- a table

Q3 needs the second fork to:

- stay the same
- change
- be doubled
- be ignored

Q4’s denominator is:

- 1
- $P(faulty)$
- 0.35
- 0.06

Lessons 92–100 revise:

- probability
- complex numbers and differential equations
- trigonometry
- integration

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $C(11, 4)$

   ANSWER 330

2. $C(6, 4)$

   ANSWER 15

3. $P(A \cap B)$ when $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$

   ANSWER 0.2

4. Are those events independent?

   ANSWER Yes — $0.5 \times 0.4 = 0.2$

5. μ of $B(12, 0.25)$

   ANSWER 3

6. σ of $B(12, 0.25)$

   ANSWER 1.5

7. z for 530 in $N(500, 15^2)$

   ANSWER 2

Medium · the standard question

7 PROBLEMS

1. P(exactly 2 women) on a committee of 4 from 6 men and 5 women

   ANSWER $\frac{150}{330} = \frac{5}{11}$

2. P(at least 1 woman) on the same committee

   ANSWER $\frac{21}{22}$

3. Bag of 7 red, 5 blue: P(same colour) without replacement

   ANSWER $\frac{31}{66}$

4. Same bag: P(second red)

   ANSWER $\frac{7}{12}$

5. Machines $65/2\backslash\%$ and $35/6\backslash\%$: P(faulty)

   ANSWER 0.034

6. Same: P(B | faulty)

   ANSWER ≈ 0.618

7. $P(X = 3)$ for $B(12, 0.25)$

   ANSWER ≈ 0.258

Hard · stretch and reason 7 PROBLEMS

1. $P(X \geq 1)$ for $B(12, 0.25)$

   ANSWER ≈ 0.968

2. Proportion of $N(500, 15^2)$ above 530

   ANSWER $\approx 2.3\backslash\%$

3. Proportion of $N(500, 15^2)$ between 485 and 515

   ANSWER $\approx 68\backslash\%$

4. A committee of 5 from 6 men and 5 women with more women than men: P

   ANSWER $\frac{181}{462} \approx 0.392$

5. Three components fail with probability 0.05 each: P at least one fails

   ANSWER ≈ 0.143

6. For $B(n, 0.25)$ with $\sigma = 3$, find n

   ANSWER $n = 48$

7. A test is $98\backslash\%$ accurate; $0.5\backslash\%$ have the disease: P(disease | positive)

   ANSWER ≈ 0.198

07 Homework

Homework — 4 tasks

Bring the decision tree to Lesson 92; complex numbers start from a blank page.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Copy the decision-tree table and add one worked example of your own to each row.
3. A bag holds 8 white and 4 black balls; three are drawn without replacement. Find the probability that all three are white and the probability that at least one is black.
4. Write your target for the revision block in one checkable sentence, and date it.

Complex numbers — arithmetic, conjugates and the Argand diagram [Cambridge revision]

One new symbol, $i^2 = -1$, and every quadratic finally has two roots.

LESSON 92–94 | 3 × 40 MIN | ALGEBRA 11, QO‘SHIMCHA BO‘LIM | P2/P3 11.1–11.4

LESSON CLOCK

17' | 26' | 26' | 26' | 21' | 4

The new number	17 min
Arithmetic	26 min
The conjugate and division	26 min
The Argand diagram	26 min
Modulus and argument	21 min
Homework	4 min

LEARNING OBJECTIVES

- Add, subtract, multiply and divide complex numbers in the form a + bi.
- Use the conjugate to divide, and to state the roots of a real quadratic.
- Plot a complex number on the Argand diagram.
- Find the modulus and argument, and convert to polar form.

TEXTBOOK

National	pp. 411–428
Cambridge	Pure Mathematics 2 & 3, pp. 246–262
Workbook	Exercise 11A–11D

01 Explanation

The new number

The equation $x^2 + 1 = 0$ has no real solution. Define one:

$$i^2 = -1$$

Then every number of the form $z = a + bi$, with a and b real, is a **complex number**: $a = Re(z)$ is the real part, $b = Im(z)$ the imaginary part. Note that $Im(z)$ is the real number b, not bi.

Z	*RE(Z)*	*IM(Z)*	KIND
$3 + 4i$	3	4	complex
$-2i$	0	-2	purely imaginary
7	7	0	real

POWERS OF *I* CYCLE WITH PERIOD FOUR

$$i^1 = i,\ i^2 = -1,\ i^3 = -i,\ i^4 = 1$$

So i^n depends only on n modulo 4: $i^{2026} = i^2 = -1$, because $2026 = 4 \times 506 + 2$.

TWO COMPLEX NUMBERS ARE EQUAL ONLY IF BOTH PARTS MATCH

$a + bi = c + di$ means $a = c$ **and** $b = d$. This turns one complex equation into two real ones — the standard technique for finding an unknown complex number.

Arithmetic

OPERATION	RULE	EXAMPLE
add	parts separately	$(3+4i) + (1-2i) = 4 + 2i$
subtract	parts separately	$(3+4i) - (1-2i) = 2 + 6i$
multiply	expand, then $i^2 = -1$	$(3+4i)(1-2i) = 11 - 2i$
square	as a binomial	$(3+4i)^2 = -7 + 24i$

The multiplication in full: $(3+4i)(1-2i) = 3 - 6i + 4i - 8i^2 = 3 - 2i + 8 = 11 - 2i$.
Nothing is new except the last step, where $-8i^2$ becomes $+8$.

EVERY QUADRATIC NOW HAS TWO ROOTS

When $D < 0$ the formula still works: $\sqrt{-16} = 4i$, so $x^2 - 2x + 5 = 0$ gives $x = \dfrac{2 \pm 4i}{2} = 1 \pm 2i$. Real quadratics with $D < 0$ have **conjugate** roots — always a pair.

The conjugate and division

$z = a + bi \Rightarrow \bar{z} = a - bi$, $z\,\bar{z} = a^2 + b^2$ — always real and non-negative

That last identity is the whole reason division works. To divide, multiply top and bottom by the conjugate of the bottom:

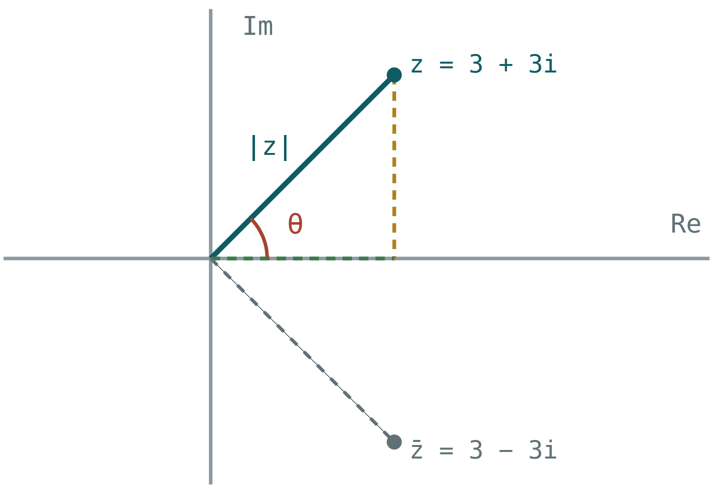
$$\frac{3 + 4i}{1 - 2i} = \frac{(3 + 4i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{-5 + 10i}{5} = -1 + 2i$$

PROPERTY	STATEMENT
sum	$z + \bar{z} = 2Re(z)$
difference	$z - \bar{z} = 2i \cdot Im(z)$
product	$z\,\bar{z} = \lvert z\rvert^2$
of a sum	$\overline{(z + w)} = \bar{z} + \bar{w}$
of a product	$\overline{(zw)} = \bar{z}\,\bar{w}$

The last two say that conjugation respects arithmetic — which is why the non-real roots of a **real** polynomial always come in conjugate pairs.

The Argand diagram

Plot $a + bi$ at the point (a, b): the horizontal axis is the real one, the vertical the imaginary. The plane of complex numbers is the **Argand diagram**.



$|z| = \sqrt{18} = 3\sqrt{2}$, $\arg z = \pi/4$

z as a point, its modulus as a distance, its argument as an angle — and $\bar{z}$ as the reflection in the real axis.

OPERATION	ON THE DIAGRAM
addition	vector addition — the parallelogram rule
conjugation	reflection in the real axis
multiplying by -1	rotation by $180°$
multiplying by i	rotation by $90°$ anticlockwise

MULTIPLYING BY I IS A QUARTER TURN

$i(a + bi) = ai + bi^2 = -b + ai$: the point (a, b) moves to $(-b, a)$, which is exactly a $90°$ rotation about the origin. This single fact makes complex numbers a tool for geometry.

Modulus and argument

$$|z| = \sqrt{a^2 + b^2}, \ arg\ z = \theta \text{ with } tan\ \theta = \frac{b}{a}$$

The modulus is the distance from the origin; the argument is the angle from the positive real axis, measured anticlockwise and normally taken in $(-\pi, \pi]$.

THE QUADRANT DECIDES THE ARGUMENT, NOT THE CALCULATOR

For $z = -1 + i$, $tan\ \theta = -1$ and a calculator returns $-45°$. But the point is in the **second** quadrant, so $arg\ z = 135° = \frac{3\pi}{4}$. Always plot first.

$$z = r(cos\ \theta + i\ sin\ \theta) \text{ — the polar form}$$

Z	$\lvert Z\rvert$	$ARG\ Z$	POLAR
$1 + i$	$\sqrt{2}$	$\frac{\pi}{4}$	$\sqrt{2}\,(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$
$-1 + i$	$\sqrt{2}$	$\frac{3\pi}{4}$	$\sqrt{2}\,(\cos\frac{3\pi}{4} + \ldots)$
$-2i$	2	$-\frac{\pi}{2}$	$2(\cos(-\frac{\pi}{2}) + \ldots)$
3	3	0	$3(\cos 0 + i\sin 0)$

WHY POLAR FORM IS WORTH THE TROUBLE

> $$|zw| = |z||w| \quad arg(zw) = arg\ z + arg\ w$$

Multiplication **multiplies** the moduli and **adds** the arguments. A product that takes four lines in Cartesian form takes one in polar form — and powers become trivial.

02 Worked examples

EXAMPLE 1 **Simplify $(3 + 4i)(1 - 2i)$ and $\frac{3 + 4i}{1 - 2i}$.**

1. $(3+4i)(1-2i) = 3 - 6i + 4i - 8i^2$
2. $= 3 - 2i + 8 = 11 - 2i$
3. $\frac{(3+4i)(1+2i)}{(1-2i)(1+2i)} = \frac{3 + 6i + 4i - 8}{5}$

   Multiply by the conjugate.
4. $= \frac{-5 + 10i}{5} = -1 + 2i$

ANSWER

$11 - 2i$ and $-1 + 2i$

EXAMPLE 2 **Solve $x^2 - 2x + 5 = 0$.**

1. $D = 4 - 20 = -16$

2. $\sqrt{-16} = 4i$

3. $x = \dfrac{2 \pm 4i}{2}$

4. $x = 1 \pm 2i$

   A conjugate pair.

ANSWER

$x = 1 + 2i$ and $x = 1 - 2i$

EXAMPLE 3 **Find $|z|$ and $arg\ z$ for $z = -1 + i$.**

1. $|z| = \sqrt{1 + 1} = \sqrt{2}$

2. $tan\ \theta = -1$

   The calculator says $-45°$.

3. The point is in the second quadrant.

4. $arg\ z = \dfrac{3\pi}{4}$

ANSWER

$|z| = \sqrt{2}$, $arg\ z = \dfrac{3\pi}{4}$

EXAMPLE 4 **Find real x and y with $(x + yi)(2 - i) = 5 + 5i$.**

1. $2x - xi + 2yi - yi^2 = (2x + y) + (2y - x)i$
2. $2x + y = 5$ and $2y - x = 5$

   Equate parts.
3. $x = 1, y = 3$
4. Check: $(1 + 3i)(2 - i) = 2 - i + 6i + 3 = 5 + 5i$.

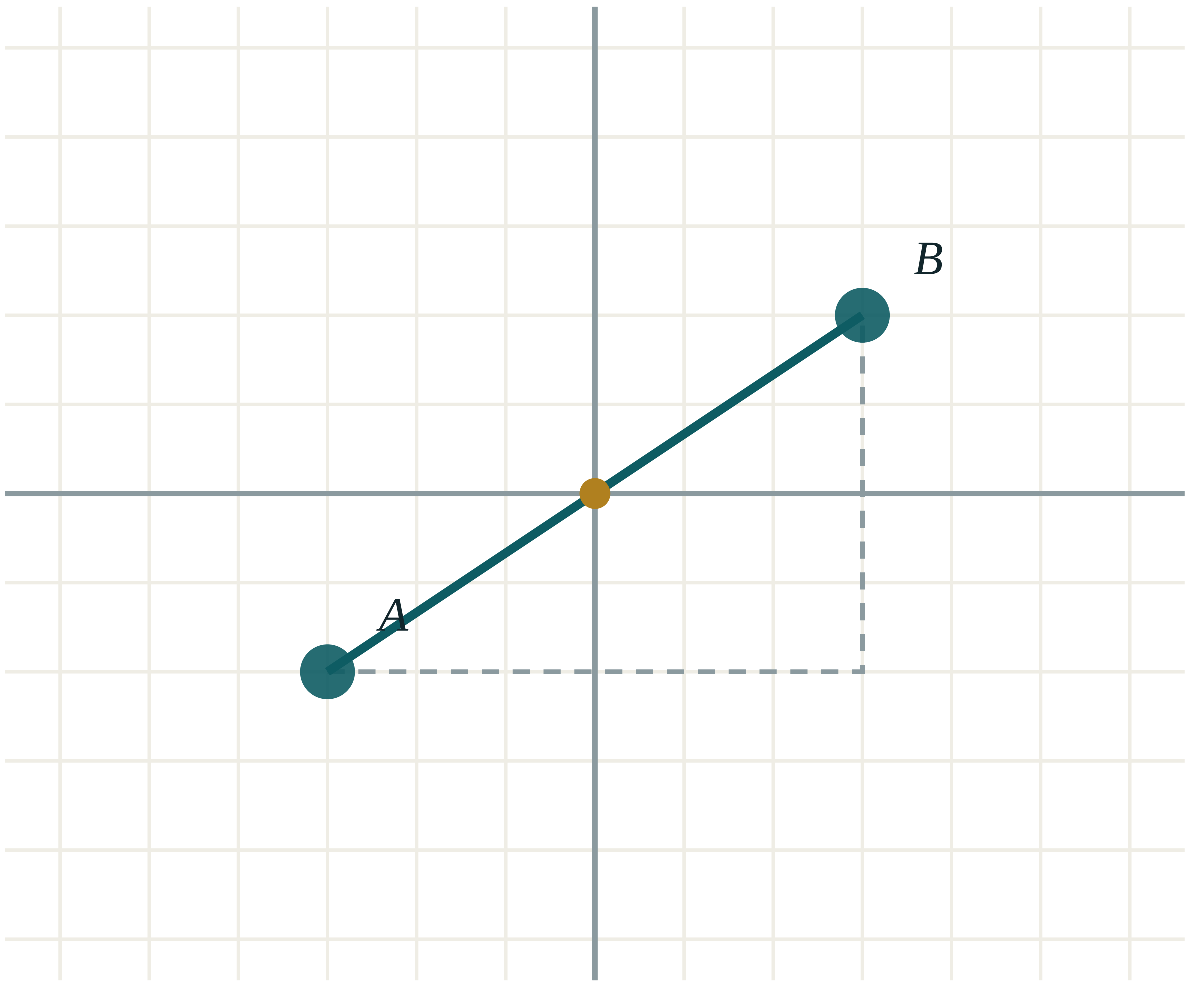
ANSWER

$x = 1, y = 3$

03 Interactive model

Draw the Argand diagram once and do every example on it — the geometry makes the algebra memorable.

The complex plane



A $(-3, -2)$ | B $(3, 2)$ | $x_2 - x_1$ **6** | $y_2 - y_1$ **4** | distance AB **7.21** | midpoint M $(0, 0)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Complex number	Kompleks son	Комплексное число
Imaginary unit	Mavhum birlik	Мнимая единица
Real part	Haqiqiy qism	Действительная часть
Imaginary part	Mavhum qism	Мнимая часть
Conjugate	Qo‘shma son	Сопряжённое число
Argand diagram	Argan diagrammasi	Диаграмма Аргана
Modulus	Modul	Модуль
Argument	Argument	Аргумент
Polar form	Trigonometrik shakl	Тригонометрическая форма
Complex plane	Kompleks tekislik	Комплексная плоскость
Purely imaginary	Sof mavhum	Чисто мнимое
Equating parts	Qismlarni tenglashtirish	Приравнивание частей

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

i^2 equals:

- 1
- -1
- i
- 0

i^{2026} equals:

- 1
- i
- -1
- $-i$

$Im(3 + 4i)$ is:

- $4i$
- 4
- 3
- 7

To divide, multiply by:

- the reciprocal
- the conjugate of the denominator
- i
- the modulus

$z\,\bar{z}$ equals:

- $|z|$
- $|z|^2$
- $2Re(z)$
- 0

Conjugation on the Argand diagram is:

- a rotation
- reflection in the real axis
- an enlargement
- a translation

Multiplying by i is:

- $90°$ anticlockwise
- $180°$
- a reflection
- a doubling

$arg(-1 + i)$ is:

- $-\frac{\pi}{4}$
- $\frac{3\pi}{4}$
- $\frac{\pi}{4}$
- $-\frac{3\pi}{4}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $(3 + 4i) + (1 - 2i)$

   ANSWER $4 + 2i$

2. $(3 + 4i) - (1 - 2i)$

   ANSWER $2 + 6i$

3. i^3

   ANSWER $-i$

4. i^4

   ANSWER 1

5. $|3 + 4i|$

   ANSWER 5

6. Conjugate of $2 - 5i$

   ANSWER $2 + 5i$

7. $Re(7)$ and $Im(7)$

   ANSWER $7, 0$

Medium · the standard question

7 PROBLEMS

1. $(3 + 4i)(1 - 2i)$

   ANSWER $11 - 2i$

2. $(3 + 4i)^2$

   ANSWER $-7 + 24i$

3. $\dfrac{3 + 4i}{1 - 2i}$

   ANSWER $-1 + 2i$

4. $\dfrac{1}{i}$

   ANSWER $-i$

5. Solve $x^2 - 2x + 5 = 0$

   ANSWER $1 \pm 2i$

6. $|-1 + i|$ and $arg(-1 + i)$

   ANSWER $\sqrt{2}, \dfrac{3\pi}{4}$

7. i^{2026}

   ANSWER -1

Hard · stretch and reason

7 PROBLEMS

1. Find real x, y with $(x + yi)(2 - i) = 5 + 5i$

   ANSWER $x = 1, y = 3$

2. Solve $z^2 = 3 + 4i$

   ANSWER $z = \pm(2 + i)$

3. Write $1 + \sqrt{3}\,i$ in polar form

   ANSWER $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$

4. Show that $(1 + i)^8 = 16$

   ANSWER $|z| = \sqrt{2}$, $\arg = \frac{\pi}{4}$, so $16(\cos 2\pi + i \sin 2\pi)$

5. One root of $x^3 - 5x^2 + 17x - 13 = 0$ is 1: find the others

   ANSWER $2 \pm 3i$

6. Prove that $z + \bar{z}$ is always real and $z - \bar{z}$ always purely imaginary

   ANSWER $2a$ and $2bi$

7. Simplify $\dfrac{(1 + i)^3}{1 - i}$

   ANSWER -2

07 Homework

Homework — 5 tasks

Plot every answer on an Argand diagram; the picture catches quadrant errors at once.

1. Simplify $(2 - 3i)(4 + i)$ and $\dfrac{2 - 3i}{4 + i}$.
2. Solve $x^2 + 4x + 13 = 0$ and plot both roots on an Argand diagram.
3. Find $|z|$ and $\arg z$ for $z = -3 - 3i$, and write z in polar form.

4. Find real x and y such that $(x + yi)(1 + 2i) = 4 - 3i$.

5. Show that i^n takes only four values, and evaluate i^{100} and i^{2027}.

Complex numbers geometrically, and loci in the complex plane [Cambridge revision]

Modulus is distance, argument is direction — and every locus becomes a picture.

LESSON 95–96 | 2 × 40 MIN | ALGEBRA 11, QO‘SHIMCHA BO‘LIM | P2/P3 11.5–11.6

LESSON CLOCK

12' | 20' | 20' | 20' | 8'

Distance and direction	12 min
The three standard loci	20 min
Regions	20 min
Greatest and least	20 min
Homework	8 min

LEARNING OBJECTIVES

- Interpret |z − a| as the distance from z to a.
- Recognise the circle, the perpendicular bisector and the half-line as loci.
- Sketch a region defined by inequalities in the complex plane.
- Find the greatest and least value of |z| on a given locus.

TEXTBOOK

National	pp. 429–440
Cambridge	Pure Mathematics 2 & 3, pp. 263–276
Workbook	Exercise 11E–11F

01 Explanation

Distance and direction

TWO READINGS THAT UNLOCK EVERYTHING

> $|z - a|$ is the **distance** from the point z to the point a

> $arg(z - a)$ is the **direction** of z as seen from a

With $a = 0$ these become the modulus and argument of z itself.

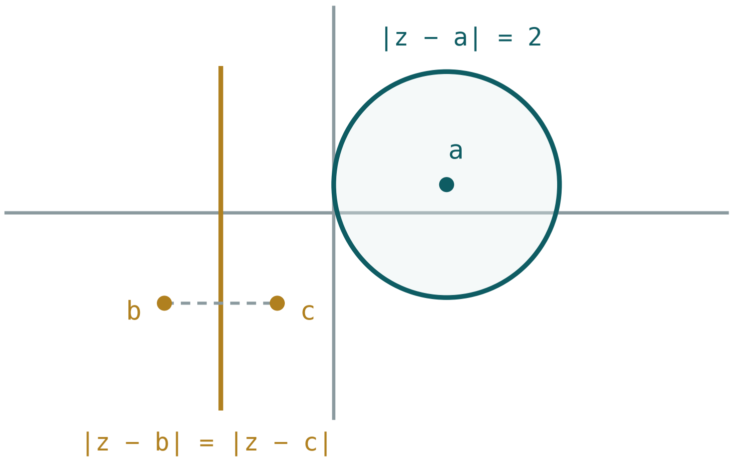
So a condition on $|z - a|$ is a condition on distance, and a condition on $arg(z - a)$ is a condition on direction. Every locus question in this lesson is one of the two.

$|Z - A|$, NOT $|Z + A|$

$|z + 3|$ is the distance to -3, not to 3 — write it as $|z - (-3)|$ before reading off the centre. This single sign is where most locus marks are lost.

The three standard loci

CONDITION	LOCUS	DESCRIBED BY
$\lvert z - a\rvert = r$	a **circle**	centre a, radius r
$\lvert z - a\rvert = \lvert z - b\rvert$	a **straight line**	the perpendicular bisector of ab
$arg(z - a) = \theta$	a **half-line**	from a (excluded) at angle θ



A circle and a perpendicular bisector — the two commonest loci, on one diagram.

Worked readings.

- $|z - 2 - i| = 3$: circle, centre $2 + i$, radius 3.
- $|z| = |z - 4|$: the perpendicular bisector of 0 and 4, that is the vertical line $Re(z) = 2$.
- $arg(z - 1) = \frac{\pi}{4}$: the half-line from 1 going up and to the right at $45°$, with 1 itself excluded (the argument of 0 is undefined).

THE HALF-LINE IS HALF A LINE

$arg(z - a) = \frac{\pi}{4}$ gives only the ray in that direction. The opposite ray has argument $-\frac{3\pi}{4}$, and belongs to a different locus. Drawing the whole line is the standard error.

Regions

Replace = by an inequality and the curve becomes a region — shaded, with the boundary solid or dashed exactly as on a number line.

CONDITION	REGION
$\lvert z - a\rvert < r$	inside the circle, boundary excluded
$\lvert z - a\rvert \geq r$	outside, boundary included
$\lvert z - a\rvert \leq \lvert z - b\rvert$	the half-plane nearer a
$0 \leq \arg(z - a) \leq \dfrac{\pi}{2}$	a quarter-plane wedge at a

Two or three conditions together give the intersection — a lens, a segment of a disc, a wedge cut by a circle. Sketch each boundary first, then shade only where all conditions hold.

TEST ONE POINT

After shading, pick an easy point — often 0 — and check it satisfies every condition. If it does and it is inside your shading, the region is right; if it does and it is outside, you have shaded the complement.

Greatest and least

"Find the greatest value of $\lvert z\rvert$ on this locus" asks for the point of the locus furthest from the origin. On a circle this needs no calculus at all.

$$\lvert a\rvert - r \leq \lvert z\rvert \leq \lvert a\rvert + r \text{ for } z \text{ on the circle } \lvert z - a\rvert = r$$

Both extremes are on the line through O and the centre: the nearest point on the near side, the furthest diametrically opposite.

LOCUS	$\lvert A\rvert$	LEAST $\lvert z\rvert$	GREATEST $\lvert z\rvert$
$\lvert z - 3 - 4i\rvert = 2$	5	3	7
$\lvert z - 6i\rvert = 2$	6	4	8
$\lvert z - 1\rvert = 3$	1	0	4

The last row: the circle passes through the origin, so the least modulus is 0 — and $\lvert a\rvert - r$ would have given -2, which is impossible. When the origin is inside or on the circle, the least value is 0.

THE TRIANGLE INEQUALITY BEHIND IT

$$\big|\,|z| - |a|\,\big| \le |z - a| \le |z| + |a|$$

Both bounds above are this inequality with $|z - a| = r$, and both are attained — which is what makes the answers exact rather than merely bounds.

02 Worked examples

EXAMPLE 1 **Describe the locus $|z - 2 - i| = 3$ and sketch it.**

1. Write it as $|z - (2 + i)| = 3$.
2. Distance from z to $2 + i$ is 3.
3. A circle, centre $(2, 1)$, radius 3.

ANSWER

A circle of centre $2 + i$ and radius 3

EXAMPLE 2 **Describe the locus $|z - 1| = |z + 3|$.**

1. $|z - 1| = |z - (-3)|$

   Both as distances.
2. Equidistant from 1 and -3.
3. The perpendicular bisector: the vertical line $Re(z) = -1$.

ANSWER

The line $Re(z) = -1$

EXAMPLE 3 **Find the greatest and least values of $|z|$ for $|z - 3 - 4i| = 2$.**

1. $|a| = |3 + 4i| = 5$
2. Least $5 - 2 = 3$.
3. Greatest $5 + 2 = 7$.

   Both on the line through the origin and the centre.

ANSWER

3 and 7

EXAMPLE 4 **Sketch the region $|z - 2i| \le 2$ and $0 \le arg\ z \le \dfrac{\pi}{2}$.**

1. Disc of centre $2i$, radius 2, boundary included.
2. Wedge from the origin between the positive real and positive imaginary axes.
3. The intersection is the right half of the disc.
4. Check $z = 1 + 2i$: in the disc, and $arg \approx 63°$ — inside.

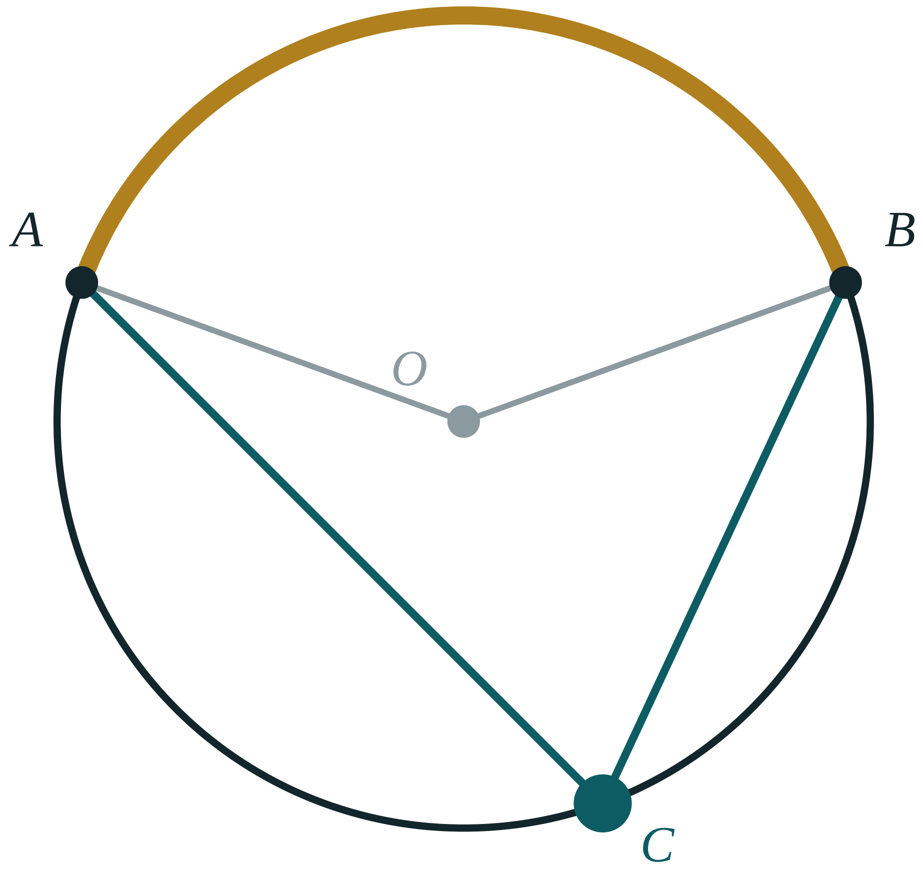
ANSWER

The right half of the disc, boundaries included

03 Interactive model

Give each pair squared paper and four loci to draw before any algebra; the descriptions then write themselves.

Circles, chords and directions



central ∠AOB **140°** | inscribed ∠ACB **70°** | ratio **2** | arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Locus	Geometrik o‘rin	Геометрическое место
Distance	Masofa	Расстояние
Circle	Aylana	Окружность
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Half-line (ray)	Nur	Луч
Region	Soha	Область
Inequality	Tengsizlik	Неравенство
Greatest value	Eng katta qiymat	Наибольшее значение
Least value	Eng kichik qiymat	Наименьшее значение
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$|z - a|$ means:

- the argument
- the distance from z to a
- the sum
- the product

$|z - a| = r$ is:

- a line
- a circle
- a half-line
- a point

$|z - a| = |z - b|$ is:

- a circle
- the perpendicular bisector of ab
- a half-line
- a point

$arg(z - a) = \theta$ is:

- a whole line
- a half-line from a
- a circle
- a wedge

$|z + 3|$ is the distance to:

- 3
- -3
- $3i$
- 0

On $|z - 3 - 4i| = 2$, the greatest $|z|$ is:

- 3
- 5
- 7
- 9

If the origin is inside the circle, the least $|z|$ is:

- $|a| - r$
- 0
- r
- $|a|$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Describe $|z| = 4$

   ANSWER Circle, centre 0, radius 4

2. Describe $|z - 2| = 3$

   ANSWER Circle, centre 2, radius 3

3. Describe $|z + 3| = 1$

   ANSWER Circle, centre -3, radius 1

4. Describe $|z| = |z - 4|$

   ANSWER The line $Re(z) = 2$

5. Describe $\arg z = \dfrac{\pi}{4}$

   ANSWER Half-line from 0 at $45°$

6. Describe $|z - i| < 2$

   ANSWER Inside the circle, boundary excluded

7. Centre of $|z - 2 - i| = 3$

   ANSWER $2 + i$

Medium · the standard question 7 PROBLEMS

1. Describe $|z - 1| = |z + 3|$

   ANSWER The line $Re(z) = -1$

2. Describe $|z - 2i| = |z - 4|$

   ANSWER The perpendicular bisector of $2i$ and 4

3. Describe $arg(z - 1) = \frac{\pi}{4}$

   ANSWER Half-line from 1, excluded

4. Greatest $|z|$ on $|z - 3 - 4i| = 2$

   ANSWER 7

5. Least $|z|$ on the same

   ANSWER 3

6. Least $|z|$ on $|z - 1| = 3$

   ANSWER 0

7. Describe $|z - 2| \leq |z|$

   ANSWER The half-plane $Re(z) \geq 1$

Hard · stretch and reason

7 PROBLEMS

1. Greatest and least $|z - 1|$ on $|z - 5 - 3i| = 2$

   ANSWER 7 and 3

2. Sketch $|z - 2i| \le 2$ and $0 \le arg\ z \le \frac{\pi}{2}$

   ANSWER The right half of the disc

3. Describe $|z - 1| = 2|z + 1|$

   ANSWER A circle — centre $-\frac{5}{3}$, radius $\frac{4}{3}$

4. Greatest $arg\ z$ on $|z - 4i| = 2$

   ANSWER $\frac{2\pi}{3}$ — at the tangent from O

5. The region $|z| \le 3$ and $|z - 3| \le 3$: describe it

   ANSWER A lens between the two circles

6. Show that $|z_1 + z_2| \le |z_1| + |z_2|$ with equality when the arguments are equal

   ANSWER The triangle inequality

7. Find the points on $|z - 3 - 4i| = 2$ of greatest and least modulus

   ANSWER $\frac{21 + 28i}{5}$ and $\frac{9 + 12i}{5}$

07 Homework

Homework — 5 tasks

Draw every locus before describing it; a sketch earns marks that words alone do not.

1. Describe and sketch $|z - 3 + 2i| = 4$, giving the centre and the radius.
2. Describe and sketch $|z - 2| = |z + 4i|$.
3. Sketch the region satisfying both $|z - 2| \le 3$ and $Im(z) \ge 0$.
4. Find the greatest and least values of $|z|$ for $|z - 5 - 12i| = 4$.

5. Explain in two sentences why $arg(z - a) = \theta$ gives a half-line and not a whole line.

Forming and solving differential equations [Cambridge revision]

An equation about a rate — and the whole of the derivative and the integral put to work.

LESSON 97–100 | 4 × 40 MIN | ALGEBRA 11, QO'SHIMCHA BO'LIM | P2/P3 10.1–10.4

LESSON CLOCK

25' | 30' | 40' | 35' | 25' | 5

What a differential equation says	25 min
Forming one	30 min
Separating the variables	40 min
Particular solutions	35 min
Interpreting	25 min
Homework	5 min

LEARNING OBJECTIVES

- Form a differential equation from a statement about a rate of change.
- Solve a separable equation, and find the general solution.
- Use an initial condition to find the particular solution.
- Interpret the solution, including its long-term behaviour.

TEXTBOOK

National	pp. 441–460
Cambridge	Pure Mathematics 2 & 3, pp. 224–245
Workbook	Exercise 10A–10D

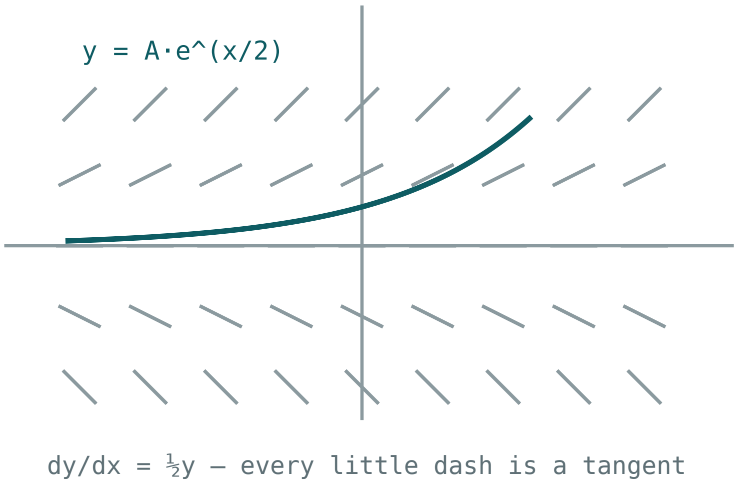
01 Explanation

What a differential equation says

An ordinary equation is a statement about a **number**. A differential equation is a statement about a **rate**, and its solution is a **function**.

$$\frac{dy}{dx} = 2x \text{ has solution } y = x^2 + c$$

Note the c: there is a whole family of solutions, one curve through each point of the plane. A single extra condition — a point the curve passes through — picks one out.



Every dash is the gradient the equation demands there; a solution is a curve that follows them.

THE DIRECTION FIELD

At each point, $\dfrac{dy}{dx}$ says what the gradient must be. Draw a short dash of that gradient at many points and the solution curves appear as the paths that follow the dashes. Two solution curves never cross — through each point runs exactly one.

Forming one

Most marks in this topic are for translating a sentence into symbols. The vocabulary is small and fixed.

ENGLISH	SYMBOLS
the rate of change of N with time	$\dfrac{dN}{dt}$
… is proportional to N	$\dfrac{dN}{dt} = kN$
… is inversely proportional to N	$\dfrac{dN}{dt} = \dfrac{k}{N}$
… decreases at a rate proportional to N	$\dfrac{dN}{dt} = -kN$
… proportional to the amount still to go	$\dfrac{dN}{dt} = k(A - N)$
… proportional to the product of N and $A - N$	$\dfrac{dN}{dt} = kN(A - N)$

PUT THE MINUS SIGN IN, AND THEN KEEP K POSITIVE

"Decreases at a rate proportional to N" is $\frac{dN}{dt} = -kN$ with $k > 0$. Writing $= kN$ and expecting k to come out negative works, but every later line then carries a sign you must remember. Declare it once.

Real cases. Radioactive decay $\frac{dN}{dt} = -kN$; Newton's law of cooling $\frac{d\theta}{dt} = -k(\theta - \theta_0)$; a population limited by resources $\frac{dP}{dt} = kP(M - P)$; a tank draining $\frac{dV}{dt} = -k\sqrt{h}$.

Separating the variables

An equation is **separable** if it can be written as a function of y times a function of x:

$$\frac{dy}{dx} = f(x)\,g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x)\,dx$$

Every equation in this course is separable. The method is three lines:

1. get all the ys with dy and all the xs with dx;
2. integrate both sides, with **one** constant;
3. rearrange for y if the question asks for it.

The standard case. $\frac{dy}{dx} = ky$:

$$\int \frac{dy}{y} = \int k\,dx \Rightarrow \ln|y| = kx + c \Rightarrow y = Ae^{kx}$$

E^C BECOMES A NEW CONSTANT A

From $\ln y = kx + c$, exponentiating gives $y = e^{kx + c} = e^c e^{kx}$. Since e^c is just some positive constant, write it as A once and never carry c again. Allowing A to be negative covers $y < 0$ too.

ONE CONSTANT, AND ADD IT AT THE INTEGRATION

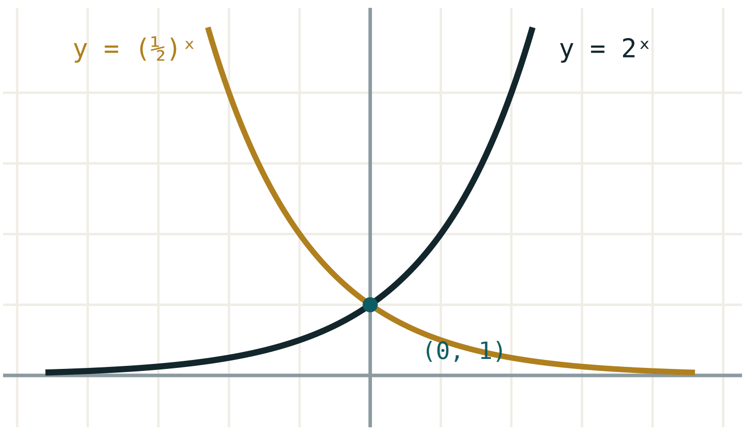
Adding c to both sides gives two constants where one will do. And adding it after rearranging — $y = Ae^{kx} + c$ — is simply a different, wrong, equation.

Particular solutions

An **initial condition** such as $y = 5$ when $x = 0$ fixes the constant. Substitute as soon as the general solution is written — before any further rearranging.

Worked case — decay. A radioactive sample obeys $\frac{dN}{dt} = -kN$, with $N = 200$ at $t = 0$ and a half-life of 5 days.

STEP	WORKING
separate	$\int \frac{dN}{N} = -\int k\,dt$
integrate	$\ln N = -kt + c$
general	$N = Ae^{-kt}$
$t = 0$	$A = 200$
half-life	$100 = 200e^{-5k} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$
particular	$N = 200e^{-0.1386t}$



both pass through (0, 1); the x-axis is an asymptote

Exponential decay — the shape of every dN/dt = −kN.

Now any question is a substitution: after 12 days, $N = 200e^{-1.663} \approx 37.9$.

Interpreting

The last mark is usually for saying what the answer **means**.

MODEL	SOLUTION	LONG TERM
$\frac{dN}{dt} = kN$	$N = Ae^{kt}$	grows without limit — unrealistic eventually
$\frac{dN}{dt} = -kN$	$N = Ae^{-kt}$	$N \to 0$
$\frac{d\theta}{dt} = -k(\theta - \theta_0)$	$\theta = \theta_0 + Ae^{-kt}$	$\theta \to \theta_0$
$\frac{dP}{dt} = kP(M - P)$	logistic	$P \to M$

EQUILIBRIUM: SET THE RATE TO ZERO

Whatever makes $\frac{dy}{dx} = 0$ is an **equilibrium** value — the solution stays there for ever if it starts there, and usually approaches it otherwise. For cooling that is $\theta = \theta_0$, room temperature; for the logistic model it is $P = M$, the carrying capacity. You can read the long-term answer off the equation without solving it.

A caution about models. Unlimited exponential growth is never a description of reality for long; it is a description of the early stage, before whatever limits growth begins to act. Saying so is often worth a mark, and is always worth saying.

02 Worked examples

EXAMPLE 1 **Solve $\frac{dy}{dx} = \frac{x}{y}$ given that $y = 3$ when $x = 0$.**

1. $\int y\,dy = \int x\,dx$

   Separate.

2. $\frac{y^2}{2} = \frac{x^2}{2} + c$

3. $9 = 0 + 2c \Rightarrow c = 4.5$

   Substitute at once.

4. $y^2 = x^2 + 9$

ANSWER

$y = \sqrt{x^2 + 9}$ (taking the positive root)

EXAMPLE 2 **A sample decays with $\frac{dN}{dt} = -kN$. It starts at 200 and has a half-life of 5 days. Find N after 12 days.**

1. $N = Ae$^{$-kt$}, $A = 200$

2. $100 = 200e$^{$-5k$} $\Rightarrow e$^{$-5k$} $= 0.5$

3. $k = \frac{\ln 2}{5} \approx 0.1386$

4. $N = 200e$^{$-0.1386(12)$} ≈ 37.9

ANSWER

≈ 38 units

EXAMPLE 3

A body at $90°$ cools in a room at $20°$. After 10 minutes it is $60°$. Find its temperature after 25 minutes.

1. $\frac{d\theta}{dt} = -k(\theta - 20) \Rightarrow \theta = 20 + Ae^{-kt}$
2. $90 = 20 + A \Rightarrow A = 70$
3. $60 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \frac{4}{7} \Rightarrow k \approx 0.05596$
4. $\theta = 20 + 70e^{-1.399} \approx 37.3°$

ANSWER

$\approx 37.3°$

EXAMPLE 4

A population satisfies $\frac{dP}{dt} = 0.02P(500 - P)$. State the equilibrium values and the long-term population.

1. Set $\frac{dP}{dt} = 0$.
2. $P = 0$ or $P = 500$.
   Two equilibria.
3. For $0 < P < 500$ the rate is positive, so P rises.
4. $P \to 500$
   The carrying capacity.

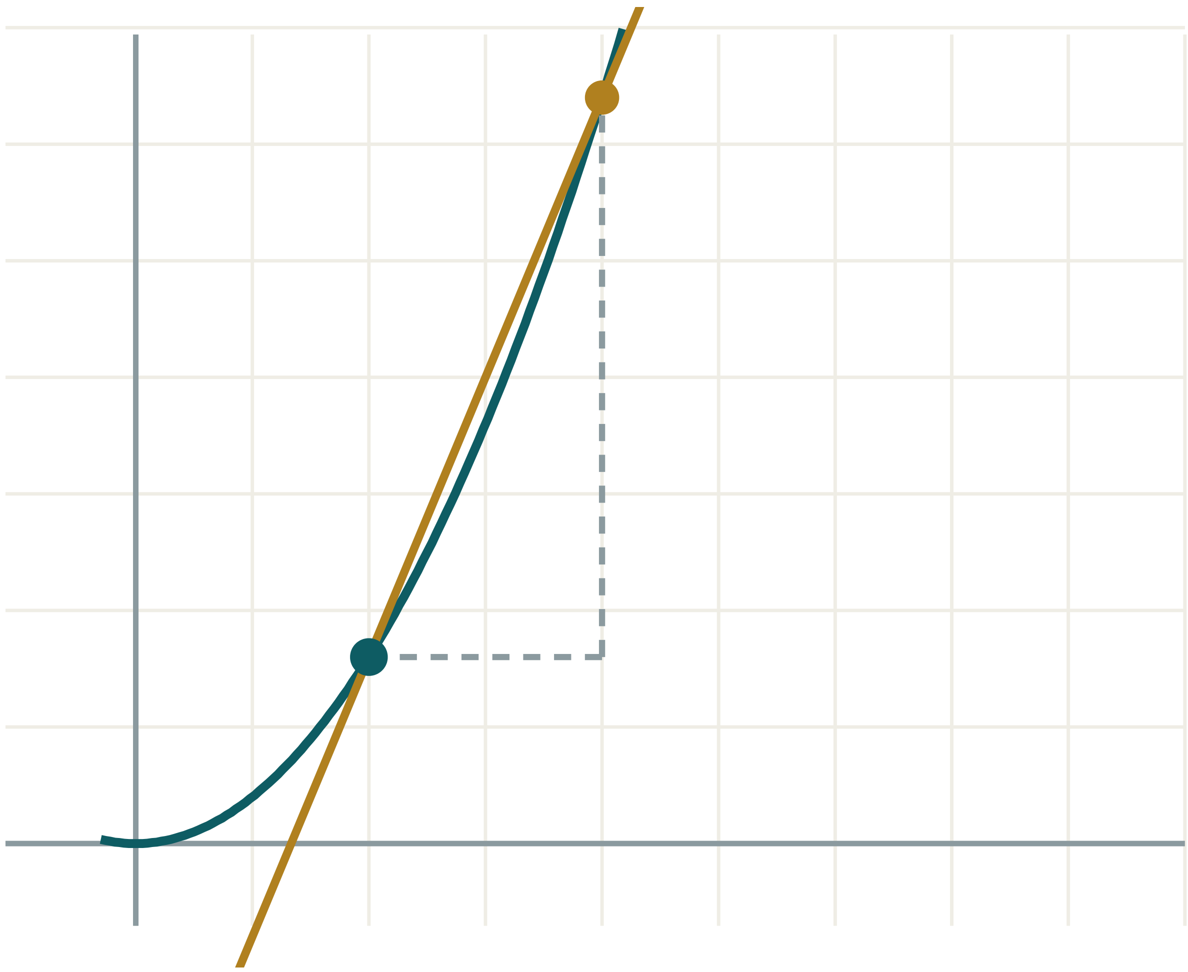
ANSWER

Equilibria 0 and 500; long term $P \to 500$

03 Interactive model

Cool a cup of hot water in front of the class, reading the thermometer every two minutes; the plotted points fall on the predicted curve.

A curve and its gradient



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Differential equation	Differensial tenglama	Дифференциальное уравнение
Rate of change	O‘zgarish tezligi	Скорость изменения
Proportional	Proporsional	Пропорциональный
Separable equation	Ajraladigan tenglama	Уравнение с разделяющимися переменными
General solution	Umumiy yechim	Общее решение
Particular solution	Xususiy yechim	Частное решение
Initial condition	Boshlang‘ich shart	Начальное условие
Arbitrary constant	Ixtiyoriy o‘zgarmas	Произвольная постоянная
Exponential growth	Eksponensial o‘sish	Экспоненциальный рост
Exponential decay	Eksponensial kamayish	Экспоненциальное убывание
Direction field	Yo‘nalishlar maydoni	Поле направлений
Equilibrium value	Muvozanat qiymati	Равновесное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The solution of a differential equation is:

- a number
- a function
- a gradient
- a constant

“Proportional to N” is:

- $\frac{dN}{dt} = kN$
- $\frac{dN}{dt} = \frac{k}{N}$
- $N = kt$
- $\frac{dN}{dt} = k$

A separable equation is solved by:

- differentiating
- separating and integrating
- squaring
- substituting

The general solution of $\frac{dy}{dx} = ky$:

- $y = kx + c$
- $y = Ae^{kx}$
- $y = \ln kx$
- $y = kx^2$

How many arbitrary constants does a first-order equation give?

- 0
- 1
- 2
- as many as you like

An equilibrium value is where:

- $y = 0$
- $\frac{dy}{dx} = 0$
- $x = 0$
- $k = 0$

For $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, the long-term temperature is:

- 0
- θ_0
- ∞
- A

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $\frac{dy}{dx} = 2x$

ANSWER $y = x^2 + c$

2. Solve $\frac{dy}{dx} = 3y$

ANSWER $y = Ae^{3x}$

3. Solve $\frac{dy}{dx} = -2y$

ANSWER $y = Ae^{-2x}$

4. Write “decreases at a rate proportional to N”

ANSWER $\frac{dN}{dt} = -kN$

5. Write “rate proportional to the amount left, $A - N$”

ANSWER $\frac{dN}{dt} = k(A - N)$

6. Equilibrium of $\frac{dy}{dt} = 3 - y$

ANSWER $y = 3$

7. Number of constants in the general solution

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = 3$

ANSWER $y = \sqrt{x^2 + 9}$

2. Solve $\frac{dy}{dx} = xy$ with $y(0) = 2$

ANSWER $y = 2e$^{$x^2/2$}

3. Solve $\frac{dy}{dx} = \frac{y}{x}$ with $y(1) = 4$

ANSWER $y = 4x$

4. Half-life 5 days, $N_0 = 200$: find k

ANSWER ≈ 0.1386

5. Same: N after 12 days

ANSWER ≈ 38

6. Cooling from $90°$ in a $20°$ room, $60°$ after 10 min: θ after 25 min

ANSWER $\approx 37.3°$

7. Equilibria of $\frac{dP}{dt} = 0.02P(500 - P)$

ANSWER 0 and 500

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{dy}{dx} = y(1 - y)$ with $y(0) = \frac{1}{2}$

   ANSWER $y = \frac{1}{1 + e^{-x}}$

2. A tank drains with $\frac{dh}{dt} = -0.2\sqrt{h}$ from $h = 25$: time to empty

   ANSWER 50 s

3. A population doubles in 8 years under $\frac{dP}{dt} = kP$: find k and the time to triple

   ANSWER $k \approx 0.0866$; ≈ 12.7 years

4. Solve $\frac{dy}{dx} = \frac{1 + y^2}{1 + x^2}$ with $y(0) = 1$

   ANSWER $y = \tan(\arctan x + \frac{\pi}{4})$

5. A drug leaves the body with $\frac{dm}{dt} = -0.15m$ from 80 mg: time to fall below 10 mg

   ANSWER ≈ 13.9 h

6. Show that $y = \theta_0 + Ae^{-kt}$ solves Newton's law of cooling

   ANSWER $\frac{d\theta}{dt} = -kAe^{-kt} = -k(\theta - \theta_0)$

7. A savings account grows at 5% continuously: time to double

   ANSWER $\frac{\ln 2}{0.05} \approx 13.9$ years

07 Homework

Homework — 5 tasks

Every answer ends with one sentence saying what the solution means in the situation described.

1. Solve $\frac{dy}{dx} = \frac{2x}{y}$ given that $y = 4$ when $x = 0$.
2. A radioactive substance has a half-life of 8 days and starts at 500 units. Form and solve the differential equation, then find the amount after 20 days.
3. A cup of tea at $95°$ cools in a room at $22°$ and reaches $70°$ after 8 minutes. Find its temperature after 20 minutes.
4. A bacterial population satisfies $\frac{dP}{dt} = 0.4P$ with $P(0) = 1000$. Find the time to reach $10\,000$, and say in one sentence why the model cannot hold for ever.
5. For $\frac{dy}{dt} = 6 - 2y$, state the equilibrium value and describe the behaviour of solutions starting above and below it.

Annual control work, and preparation for the entrance paper

Two years of calculus in one paper, then what the university examination actually asks.

LESSON 101–102 | 2 × 40 MIN | ALGEBRA 11, YILLIK NAZORAT ISHI | ANNUAL REVIEW

LESSON CLOCK

Instructions	3 min
The paper	50 min
Answers	12 min
Diagnosis	10 min
The entrance paper	5 min

LEARNING OBJECTIVES

- Apply differentiation, integration, probability and complex numbers in one paper.
- Choose the shortest route to each answer under time pressure.
- Name each lost mark and rewrite the solution in full.
- Recognise the standard question types of the entrance examination.

TEXTBOOK

National	pp. 461–464
Cambridge	Pure Mathematics 2 & 3, pp. 277–280
Workbook	Annual paper A

01 Explanation

The paper — 50 marks, 50 minutes

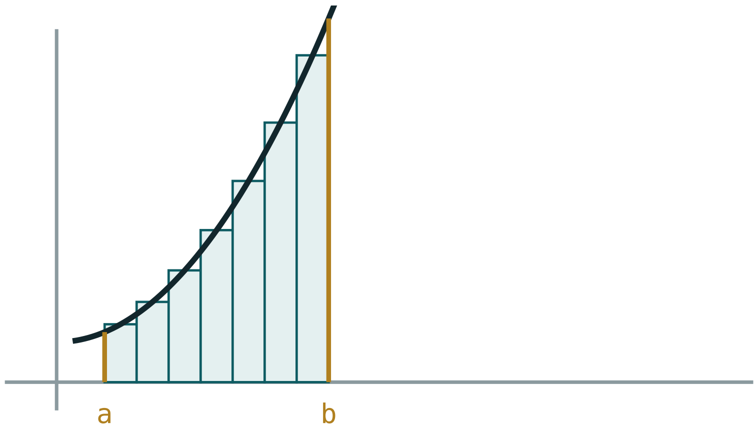
Q	TASK	MARKS	FROM
1	Differentiate $y = x^3 \ln x$ and $y = \dfrac{\sin x}{x^2}$	*6*	Q I
2	Find the stationary points of $y = x^3 - 6x^2 + 9x + 2$ and classify them	*7*	Q I
3	Evaluate $\int_1^4 (2x + \dfrac{1}{\sqrt{x}})\, dx$ and find the area between $y = x^2$ and $y = 2x$	*8*	Q II–III
4	A box of 4 red and 6 blue: two drawn without replacement — the tree and three probabilities	*7*	Q IV
5	$X \sim B(10, 0.3)$: μ, σ, $P(X = 2)$, $P(X \geq 1)$	*7*	Q IV
6	Simplify $\dfrac{2 + 3i}{1 - i}$, solve $z^2 + 2z + 5 = 0$ and describe $\lvert z - 2i \rvert = 3$	*8*	Q IV
7	A sample decays with half-life *6* days from *400* units: form and solve the equation, then find $N(15)$	*7*	Q IV

HOW TO SPEND 50 MINUTES

Questions 1, 5 and 6 are mechanical — about *18* minutes for *21* marks. Do them first. Questions 2, 3 and 7 reward care rather than speed. Leave whichever of them you find hardest to last, and write the setting-up even if you cannot finish: method marks are real marks.

The two years in one page

BLOCK	THE ONE IDEA	THE ONE FORMULA
the derivative	the gradient of the tangent	$\frac{dy}{dx} = \lim \frac{\Delta y}{\Delta x}$
applications	$f' = 0$ locates the turning points	f'' *decides the kind*
the integral	the reverse of the derivative	$\int f' = f + c$
the definite integral	signed area	$\int_a^b f = F(b) - F(a)$
probability	count, or measure, then divide	$P = \frac{m}{n}$
distributions	the shape of many repetitions	$\mu = np$
complex numbers	$i^2 = -1$	$z\bar{z} = \lvert z\rvert^2$
differential equations	a statement about a rate	*separate and integrate*



more, thinner rectangles → the definite integral

The picture that connects the two halves of the course — the integral as area.

The entrance paper

The university entrance examination in mathematics rewards accuracy and speed on standard types rather than ingenuity. The types repeat.

TYPE	WHAT IT ASKS	WHERE IT CAME FROM
algebraic	simplify, solve, factorise	Grades 8–10
function	domain, range, graph, inverse	Grade 10 Q I
trigonometric	equations and identities	Grade 10 Q IV
exponential and log	equations and inequalities	Grade 10 Q III
calculus	tangent, extremum, area	Grade 11 Q I–III
combinatorial	counting and probability	Grade 11 Q III–IV
geometric	a solid, or a coordinate figure	Grade 10–11 geometry

FOUR RULES THAT RAISE A SCORE WITHOUT NEW MATHEMATICS

- **Read to the end** before starting; the last clause often changes the method.
- **Write the setting-up** even when stuck — a, d, n, p, the sketch, the formula. Method marks do not require an answer.
- **Check the domain**: a root, a logarithm, a denominator, $|t| \le 1$ — every rejected root is a mark.
- **Leave nothing blank**. A named formula and one substituted line is worth more than white space, and costs twenty seconds.

What to practise over the summer. Twenty past papers, timed, one a day, marked the same evening against the scheme — and a written list of every type that cost a mark twice. That list, not the papers, is what improves the score.

02 Worked examples

EXAMPLE 1 Model answer, Q2: the stationary points of $y = x^3 - 6x^2 + 9x + 2$.

1. $y' = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$
2. $x = 1$ and $x = 3$.
3. $y'' = 6x - 12$: $y''(1) = -6 < 0$, $y''(3) = 6 > 0$.
4. Maximum $(1, 6)$, minimum $(3, 2)$.

ANSWER

Maximum $(1, 6)$, minimum $(3, 2)$

EXAMPLE 2 Model answer, Q3: the area between $y = x^2$ and $y = 2x$.

1. $x^2 = 2x \Rightarrow x = 0, 2$
2. $\int_0^2 (2x - x^2)\,dx$

   Upper minus lower.
3. $= \left[x^2 - \frac{x^3}{3}\right]_0^2$
4. $= 4 - \frac{8}{3} = \frac{4}{3}$

ANSWER

$\frac{4}{3}$

EXAMPLE 3 **Model answer, Q6: simplify $\frac{2 + 3i}{1 - i}$ and solve $z^2 + 2z + 5 = 0$.**

1. $$\frac{(2 + 3i)(1 + i)}{2} = \frac{2 + 2i + 3i - 3}{2}$$

2. $$= \frac{-1 + 5i}{2}$$

3. $D = 4 - 20 = -16 \Rightarrow \sqrt{D} = 4i$

4. $z = -1 \pm 2i$

ANSWER

$\frac{-1 + 5i}{2}$ and $z = -1 \pm 2i$

EXAMPLE 4 **Model answer, Q7: half-life *6* days from *400* units — find *N(15)*.**

1. $\frac{dN}{dt} = -kN \Rightarrow N = 400e^{-kt}$

2. $200 = 400e^{-6k} \Rightarrow k = \frac{\ln 2}{6} \approx 0.1155$

3. $N(15) = 400e^{-1.733}$

4. ≈ 70.7

ANSWER

≈ 71 units

03 Interactive model

Hand out the mark allocation before the paper starts; ten seconds of planning is worth five marks, every time.

The two years in twelve questions

$\frac{d}{dx}(x^3 \ln x)$ is:

- $3x^2 \ln x$
- $3x^2 \ln x + x^2$
- x^2
- $3x^2 + \frac{1}{x}$

A maximum has:

- $f'' > 0$
- $f'' < 0$
- $f' > 0$
- $f'' = 0$

$\int x^2\,dx$ is:

- $2x + c$
- $\frac{x^3}{3} + c$
- $x^3 + c$
- $\frac{x^3}{2} + c$

$\int_0^2 2x\,dx$ is:

- 2
- 4
- 8
- 0

Area between curves is:

- upper minus lower
- lower minus upper
- their sum
- their product

$P(A \cap B)$ in general:

- $P(A)P(B)$
- $P(A)P(B \mid A)$
- $P(A) + P(B)$
- 0

The mean of $B(10, 0.3)$:

- 3
- 7
- 2.1
- 10

Within two σ lies:

- 68%
- 95%
- 99.7%
- 50%

i^2 is:

- 1
- -1
- i
- 0

$|z - a| = r$ is:

- a line
- a circle
- a half-line
- a point

The solution of $\frac{dy}{dx} = ky$:

- $y = kx + c$
- $y = Ae^{kx}$
- $y = \ln kx$
- $y = kx^2$

Equilibrium is where:

- $y = 0$
- $\frac{dy}{dx} = 0$
- $x = 0$
- $k = 0$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Derivative	Hosila	Производная
Integral	Integral	Интеграл
Stationary point	Statsionar nuqta	Стационарная точка
Definite integral	Aniq integral	Определённый интеграл
Probability	Ehtimollik	Вероятность
Complex number	Kompleks son	Комплексное число
Entrance examination	Kirish imtihoni	Вступительный экзамен
Time management	Vaqtni taqsimlash	Распределение времени
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which questions should be done first?

- in order
- the mechanical ones — 1, 5 and 6
- the hardest
- the longest

A blank answer earns:

- nothing
- a method mark for the formula
- half marks
- a warning

The entrance paper rewards:

- ingenuity
- accuracy and speed on standard types
- long proofs
- guessing

The best summer practice is:

- reading
- timed past papers, marked the same evening
- new topics
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\frac{d}{dx}(x^3 \ln x)$

ANSWER $3x^2 \ln x + x^2$

2. $\frac{d}{dx}\left(\frac{\sin x}{x^2}\right)$

ANSWER $\frac{x \cos x - 2 \sin x}{x^3}$

3. $\int_0^2 2x\,dx$

ANSWER 4

4. μ of $B(10, 0.3)$

ANSWER 3

5. σ of $B(10, 0.3)$

ANSWER ≈ 1.45

6. Solve $z^2 + 2z + 5 = 0$

ANSWER $-1 \pm 2i$

7. Describe $|z - 2i| = 3$

ANSWER Circle, centre $2i$, radius 3

Medium · the standard question

7 PROBLEMS

1. Stationary points of $y = x^3 - 6x^2 + 9x + 2$

   ANSWER $(1, 6)$ max, $(3, 2)$ min

2. $\int_1^4 (2x + \frac{1}{\sqrt{x}})\, dx$

   ANSWER $[x^2 + 2\sqrt{x}]_1^4 = 20 - 3 = 17$

3. Area between $y = x^2$ and $y = 2x$

   ANSWER $\frac{4}{3}$

4. Box of 4 red, 6 blue, two drawn without replacement: $P(RR)$

   ANSWER $\frac{2}{15}$

5. Same: P(one of each)

   ANSWER $\frac{8}{15}$

6. $P(X = 2)$ for $B(10, 0.3)$

   ANSWER ≈ 0.233

7. Simplify $\frac{2 + 3i}{1 - i}$

   ANSWER $\frac{-1 + 5i}{2}$

Hard · stretch and reason

7 PROBLEMS

1. $P(X \geq 1)$ for $B(10, 0.3)$

   ANSWER ≈ 0.972

2. Half-life 6 days from 400: $N(15)$

   ANSWER ≈ 71

3. Greatest and least $|z|$ on $|z - 2i| = 3$

   ANSWER 5 and 0 — the origin is inside

4. Volume when $y = x^2$ from 0 to 2 rotates about Ox

   ANSWER $\frac{32\pi}{5}$

5. A box of 4 red and 6 blue: P(second red)

   ANSWER $\frac{2}{5}$

6. Show that $y = x^3 - 6x^2 + 9x + 2$ has an inflection at $x = 2$

   ANSWER $y'' = 6x - 12 = 0$

7. Solve $\frac{dy}{dx} = \frac{y}{x^2}$ with $y(1) = e$

   ANSWER $y = e^2e$^{−1/x}

07 Homework

Homework — 4 tasks

This is the last algebra lesson of the school course. The summer list is the part that matters.

1. Rewrite in full every annual-paper question that lost a mark, naming the slip in the margin.
2. Write the two years in one page: the eight blocks with one formula and one sketch each.
3. Work one complete past entrance paper under timed conditions and mark it against the scheme the same evening.

4. Write the list of question types that cost you a mark more than once, and one sentence saying how you will practise each.